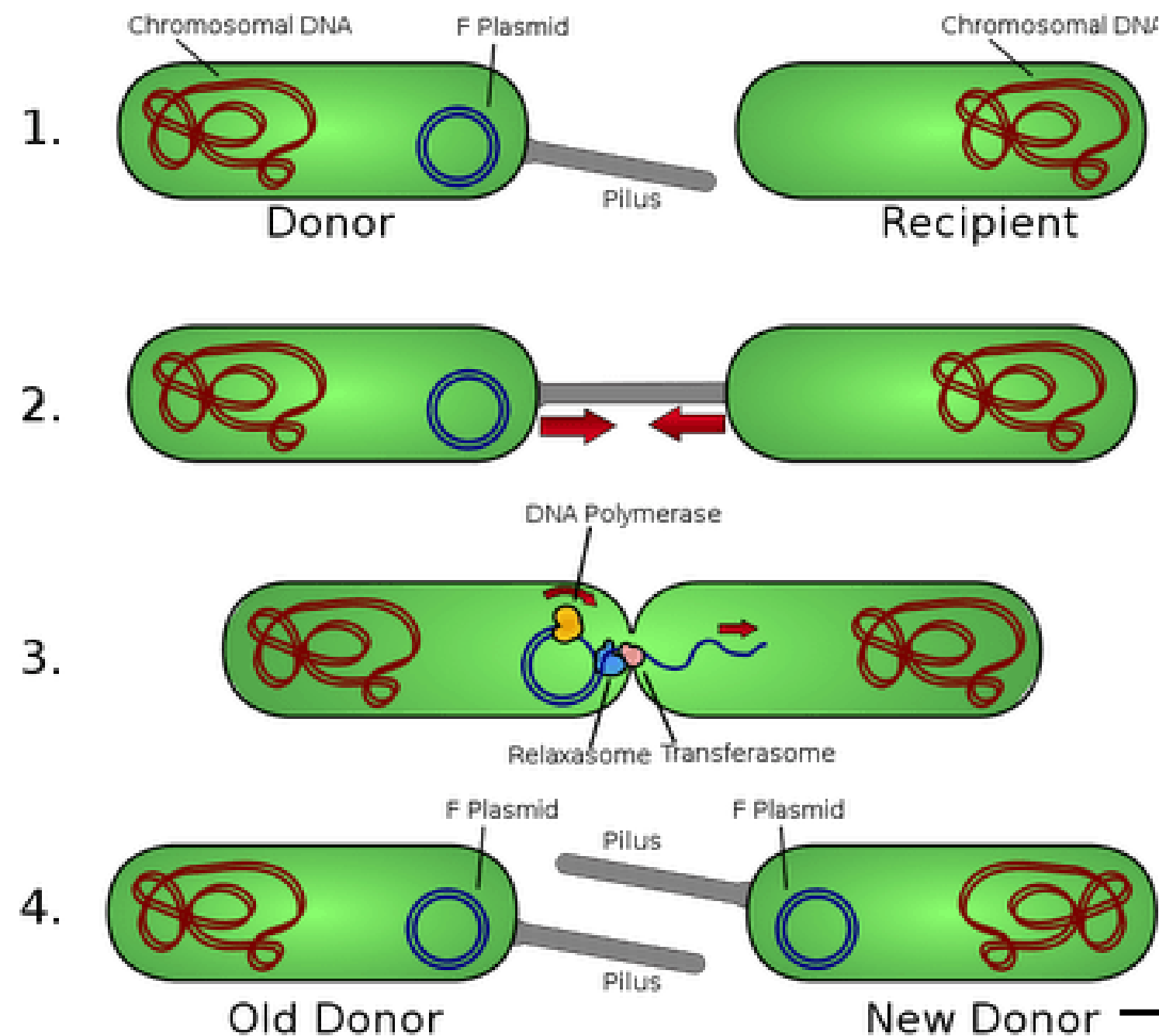


Incontro 9/09

Tesi Magistrale - Stefano Ferri

- Fattori che influenzano l'esito del **trasferimento genico orizzontale** in *Escherichia coli*: uno studio di caso utilizzando i ceppi CL218 e J53.



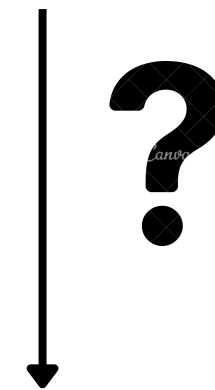
Coniugazione batterica

Donatore: CL218 resistente al meropenem

Ricevente: J53 non resistente al meropenem

- **Fattori** che influenzano l'esito del trasferimento genico orizzontale in *Escherichia coli*: uno studio di caso utilizzando i ceppi CL218 e J53.

- Temperatura
 - Concentrazione di Ibuprofene
 - Concentrazione di dimetilsolfossido (DMSO)
- Effetto su trasferimento plasmidi
- } Effetti su crescita transcongiuganti, permeabilità membrana cellulare, assorbimento



Caratterizzare questi effetti e valutare interazioni tra fattori

- Fattori che influenzano **l'esito** del trasferimento genico orizzontale in *Escherichia coli*: uno studio di caso utilizzando i ceppi CL218 e J53.

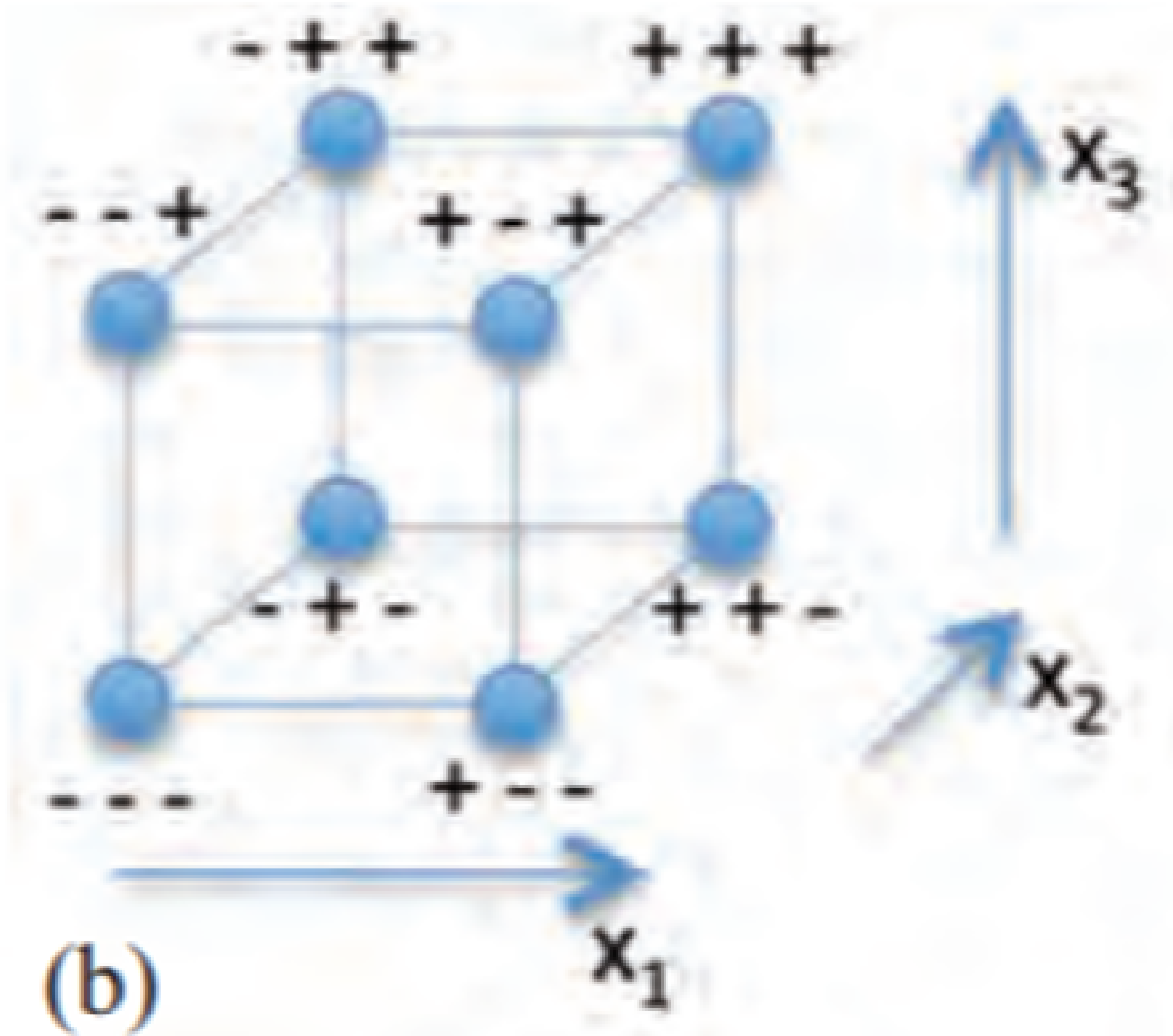


- Crescita transcongiuganti ("Conta batterica")
- Frequenza di coniugazione
transcongiuganti / # donatori
- Trasferimento plasmidi (inc N, inc F)

Piano sperimentale - DOE fattoriale completo

Fattori	-	+
T [°C]	28	40
Ibuprofene [mg/l]	6,25	50
DMSO [%v/v]	0	10

# Esperimento	T [°C]	Ibuprofene [mg/l]	DMSO
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-
5	-	+	+
6	-	-	+
7	+	-	+
8	+	+	+

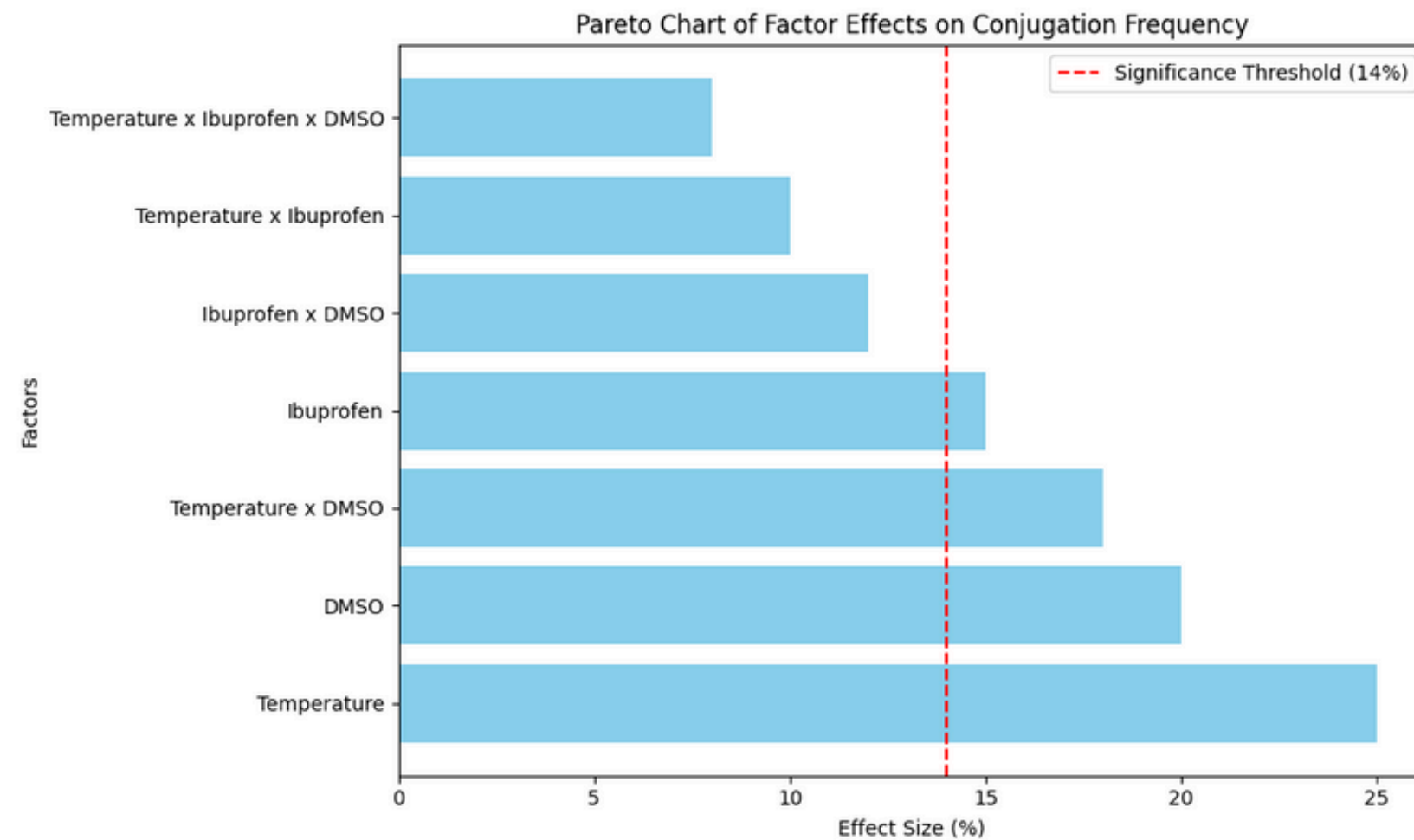


+

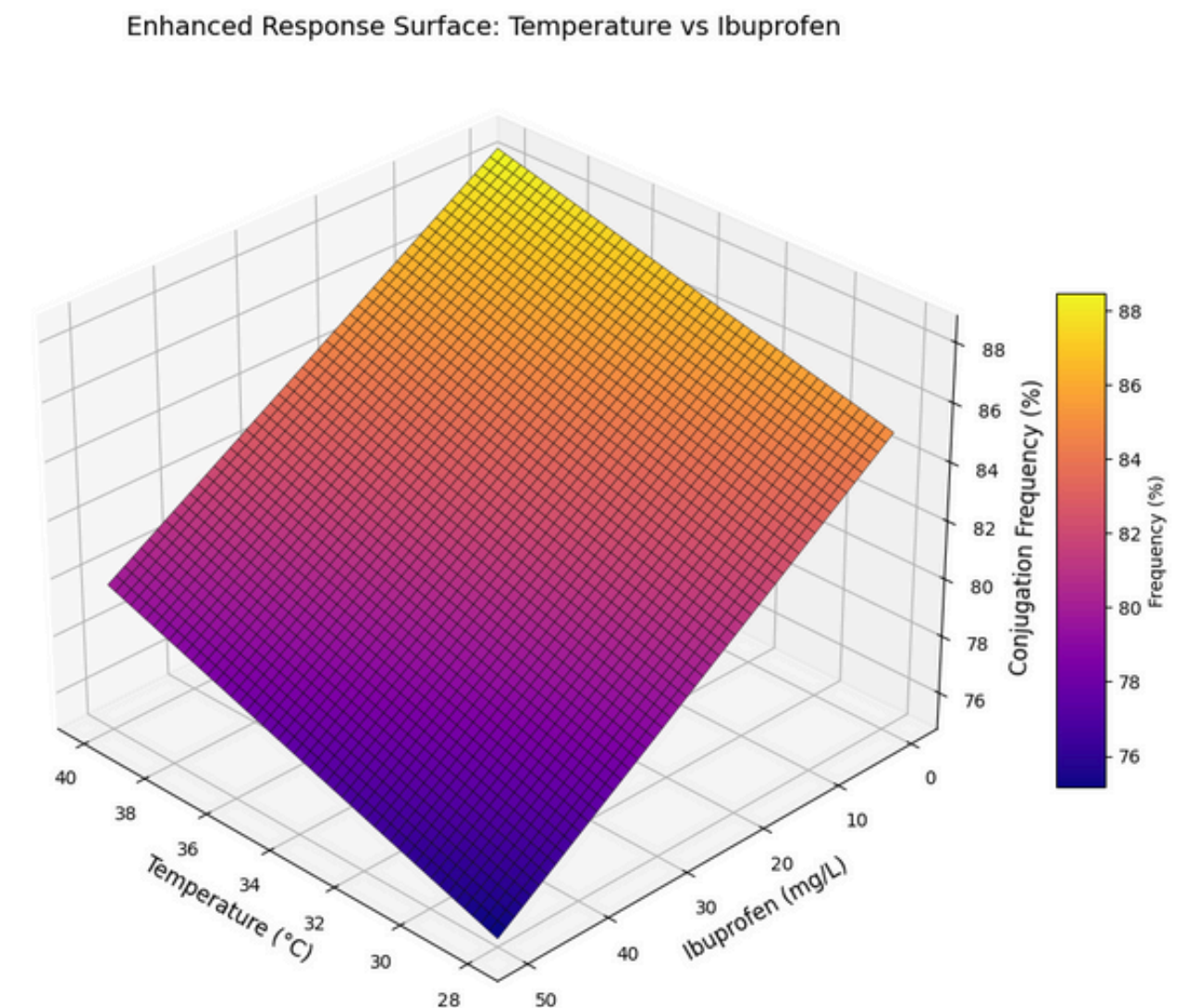
# Esperimento	T [°C]	Ibuprofene [mg/l]	DMSO
9	-	0	+
10	+	0	+

Elaborazioni DOE

Frontiera di Pareto



Superficie di risposta

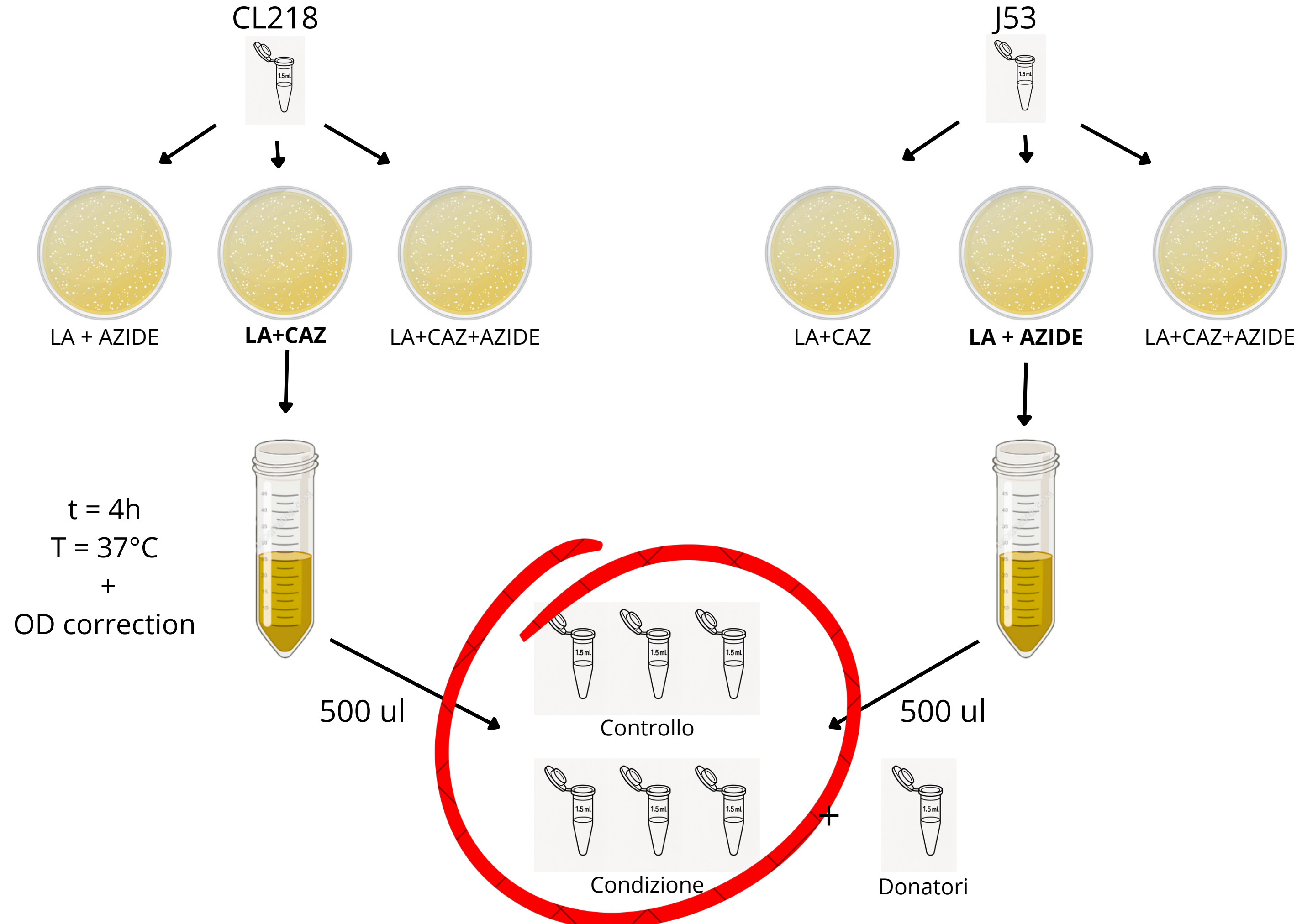


$$y = b_0 + b_1(x_1) + b_2(x_2) + b_3(x_3) + b_{12}(x_1)(x_2) + b_{13}(x_1)(x_3) + b_{23}(x_2)(x_3) + b_{123}(x_1)(x_2)(x_3) + \epsilon$$

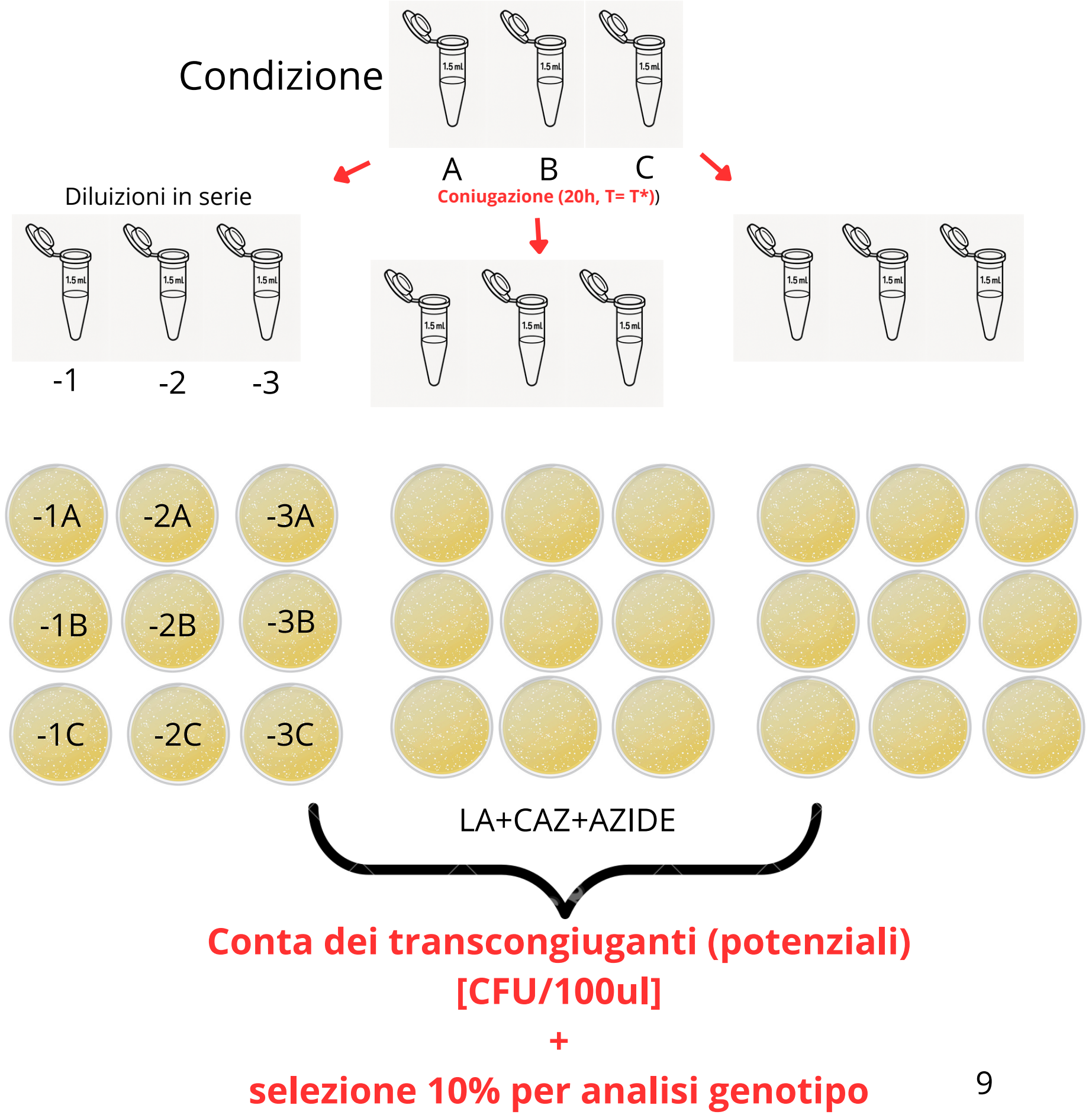
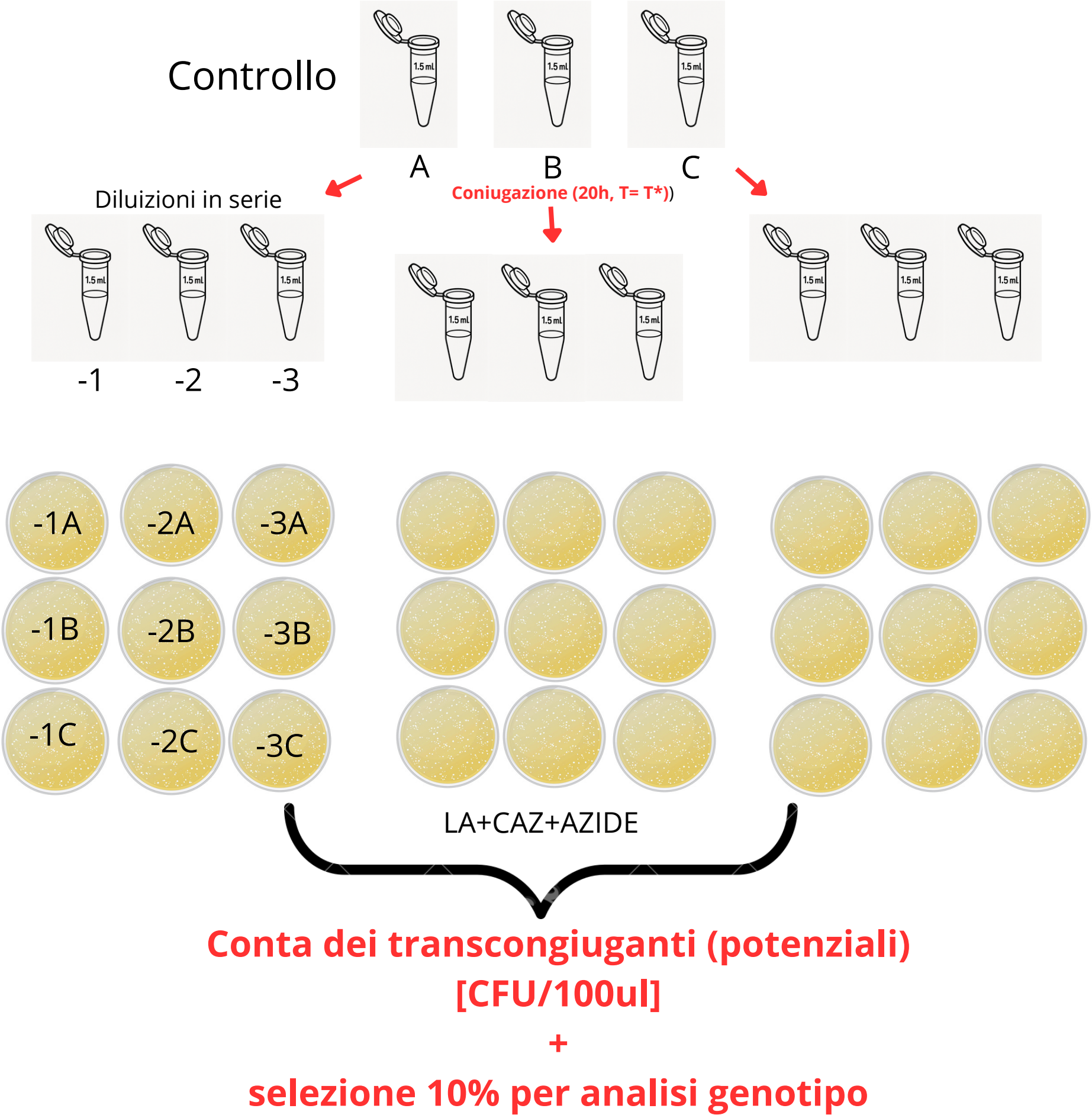
Obiettivi sperimentali

1. **Recupero dei potenziali transconiuganti su terreno contenente azide e ceftazidime (CAZ).**
2. **Genotipizzazione per confermare il genotipo del ceppo ricevente.**
3. **Conferma della presenza di ciascun gene: incN (plasmide che ospita il gene blaKPC), incF.**
4. **Calcolo della frequenza di coniugazione.**

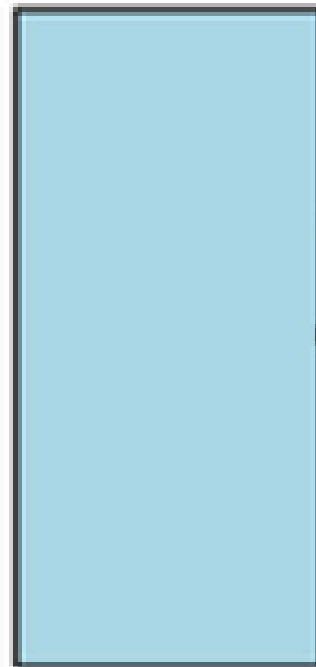
1- Recupero dei potenziali transconiuganti su terreno contenente azide e ceftazidime (CAZ).



1- Recupero dei potenziali transconiuganti su terreno contenente azide e ceftazidime (CAZ).

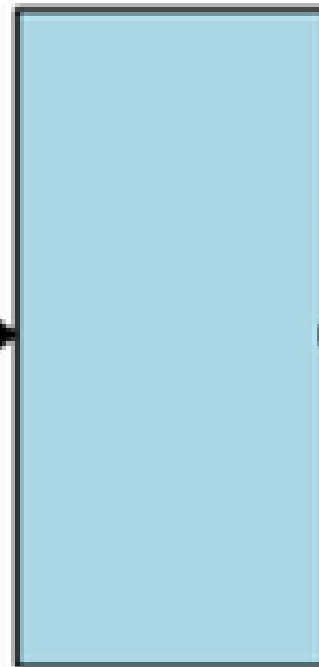


Soluzione madre



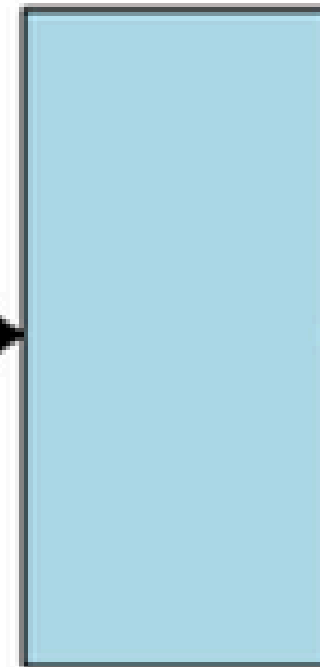
100 μ L

Diluizione -1



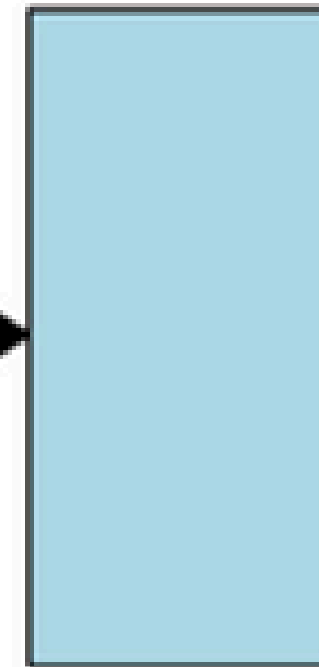
100 μ L

Diluizione -2



100 μ L

Diluizione -3

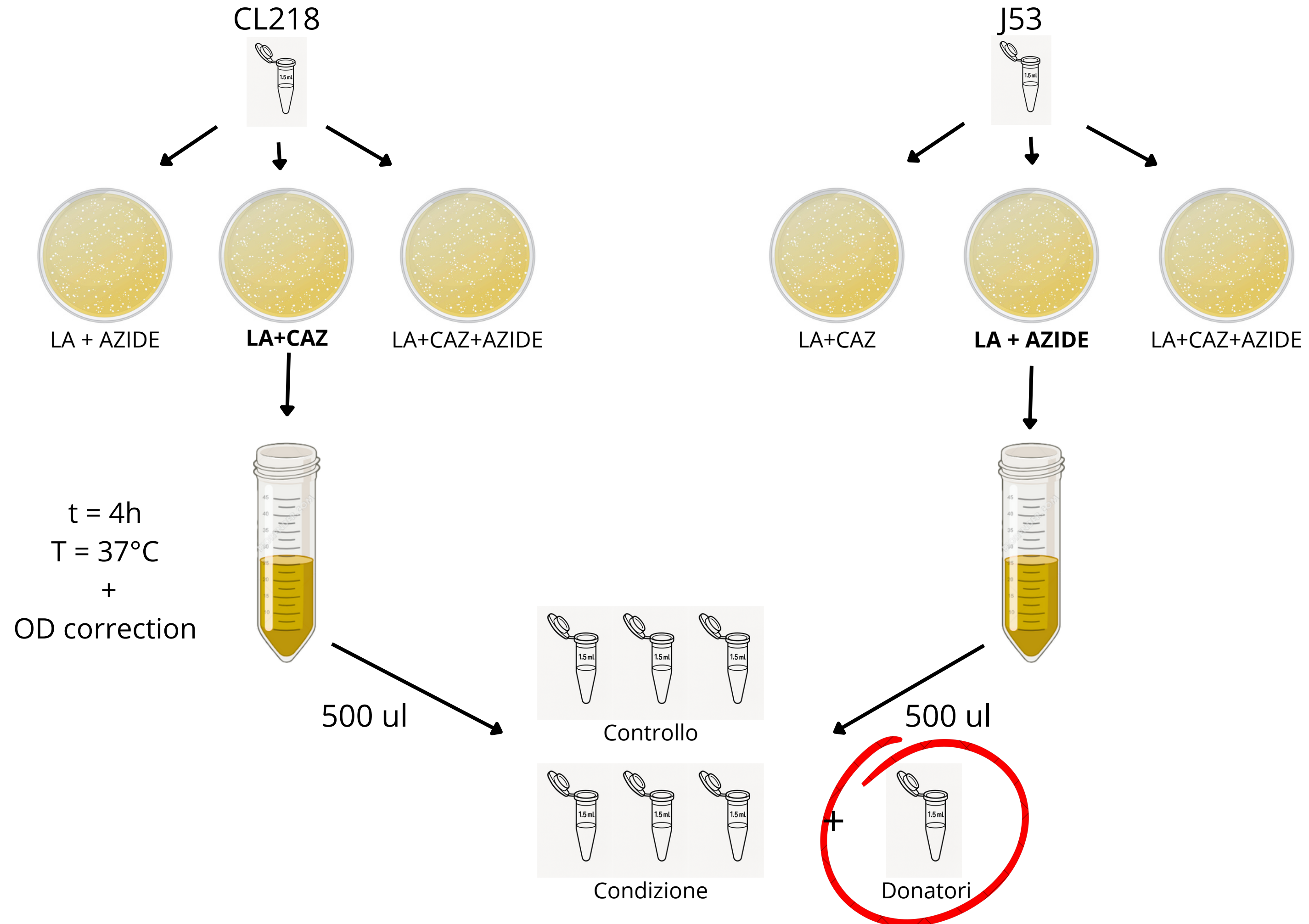


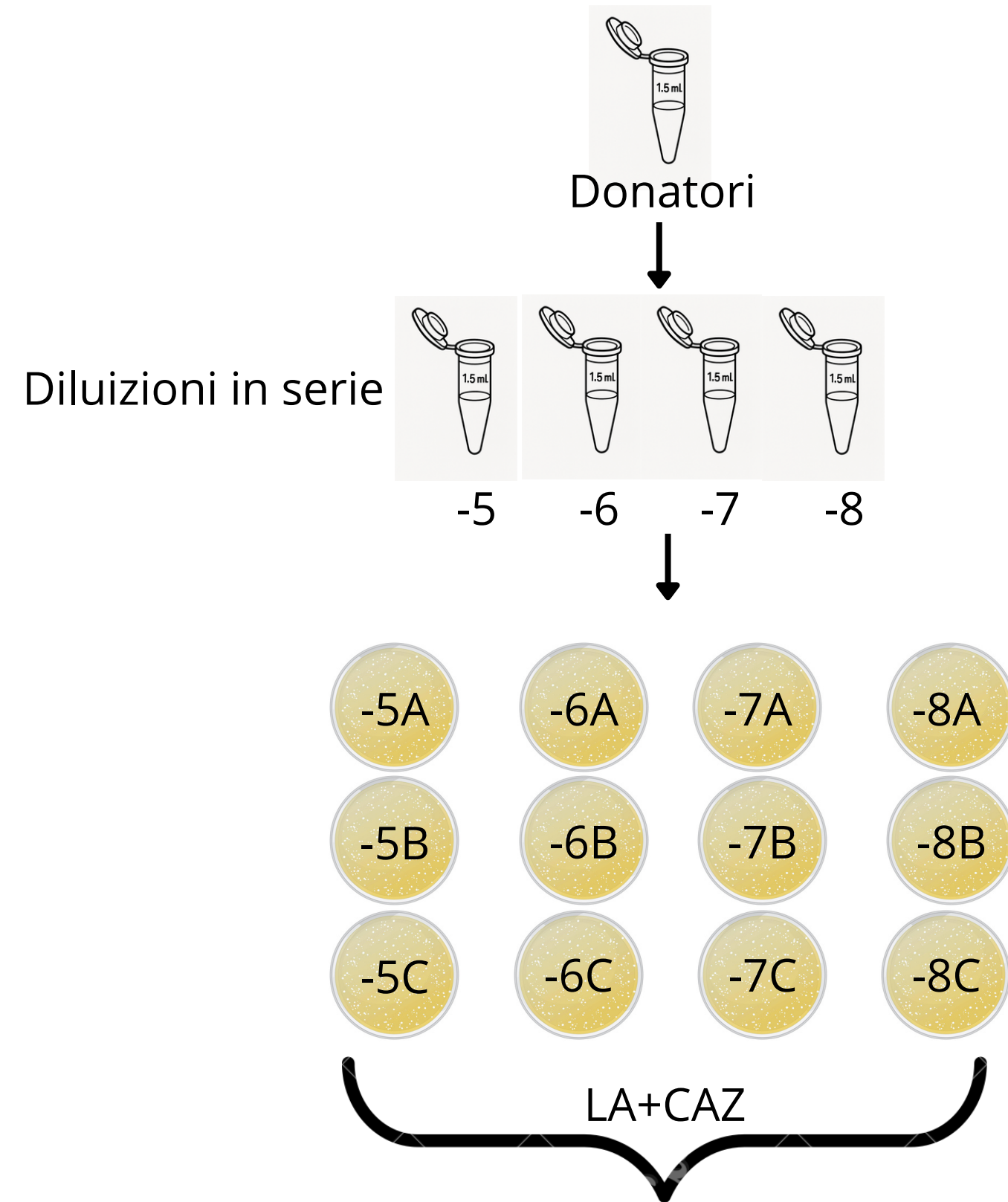
+ 900 μ L diluente

+ 900 μ L diluente

+ 900 μ L diluente

1- Recupero dei potenziali transconiuganti su terreno contenente azide e ceftazidime (CAZ).





Conta dei donatori [CFU/100ul]

donatori

3- Genotipizzazione per confermare il genotipo del ceppo ricevente.

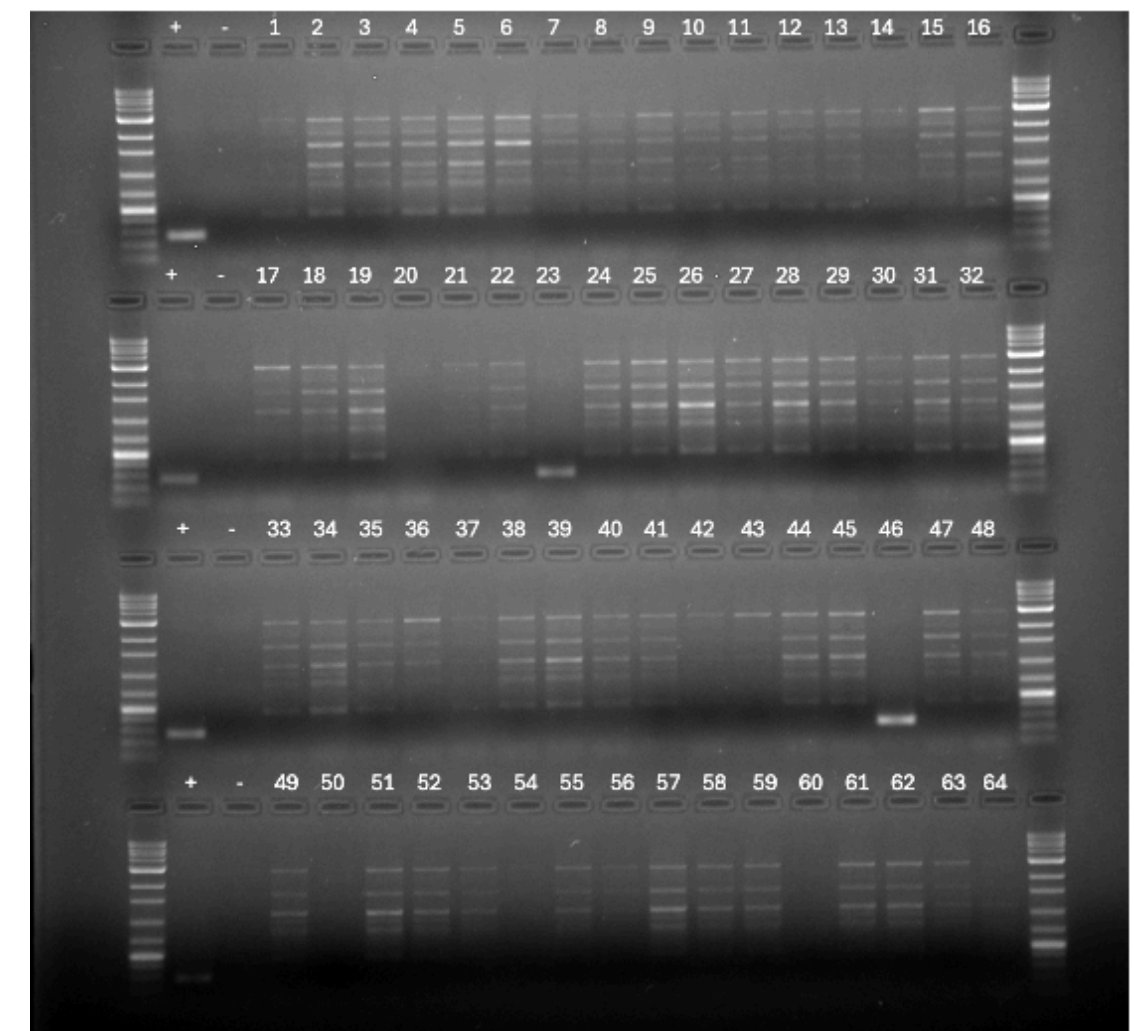
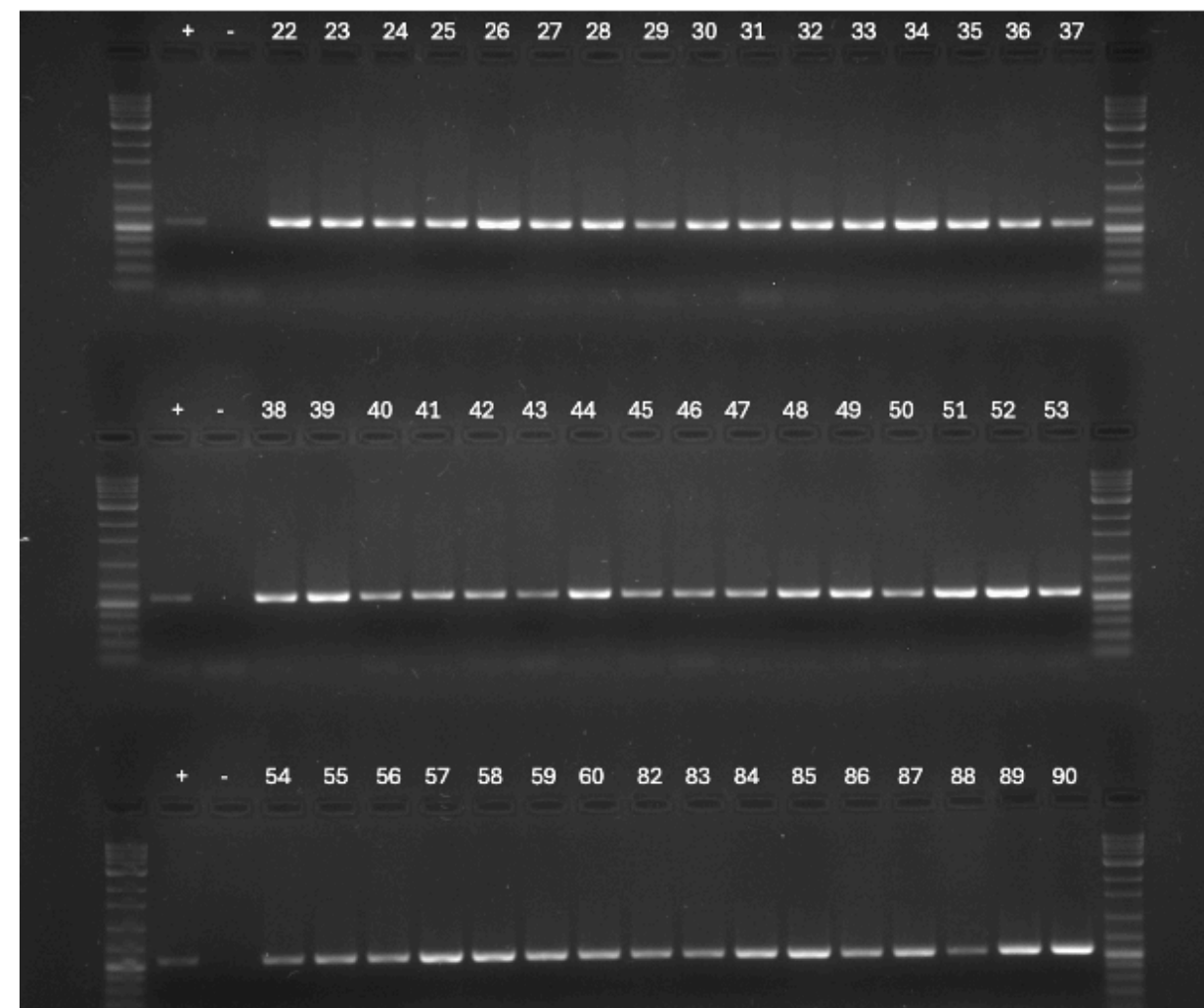
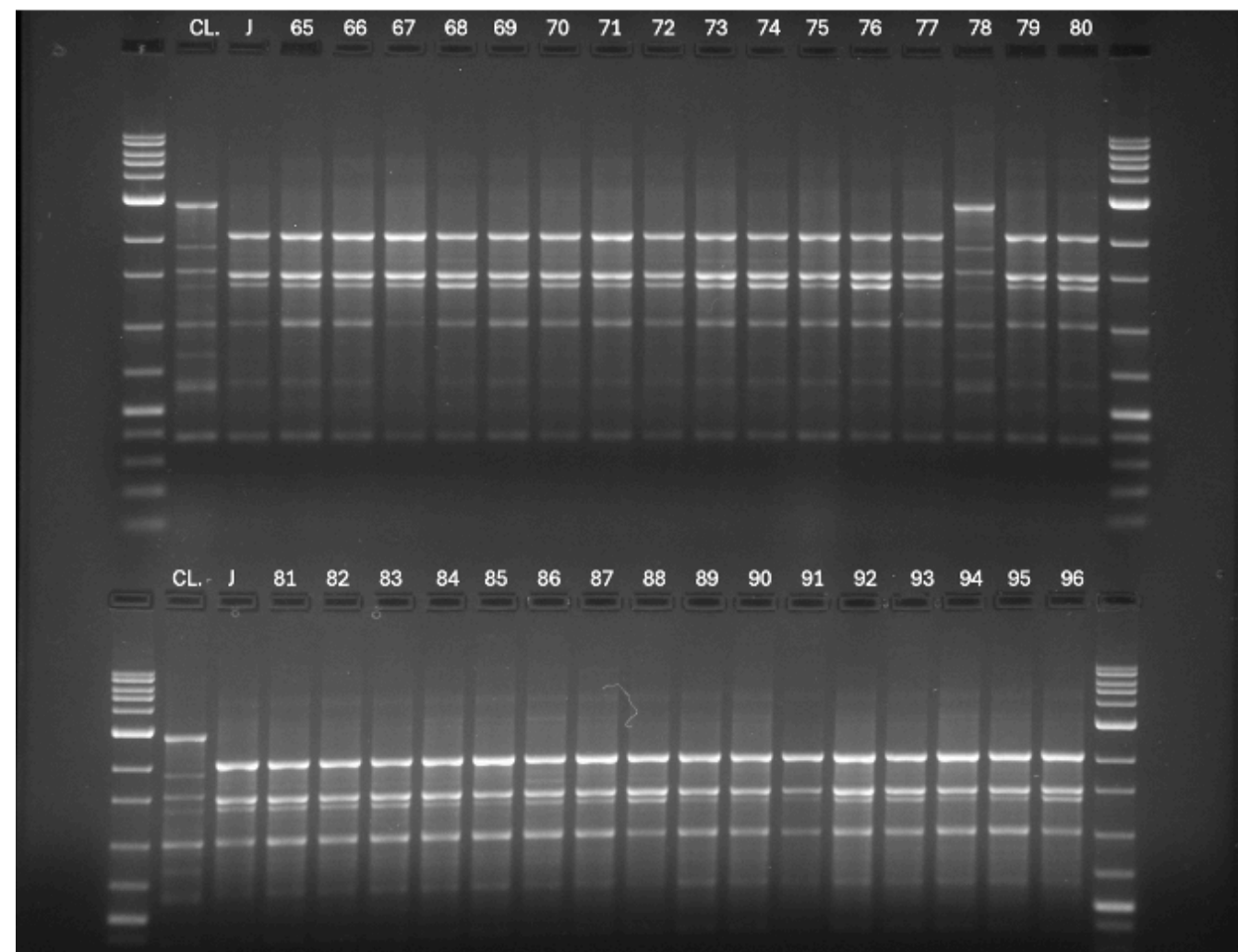
4- Conferma della presenza di ciascun gene: *incN* (plasmide che ospita il gene *blaKPC*), *incF*.

Veri transcongiuganti

RAPD

Inc N

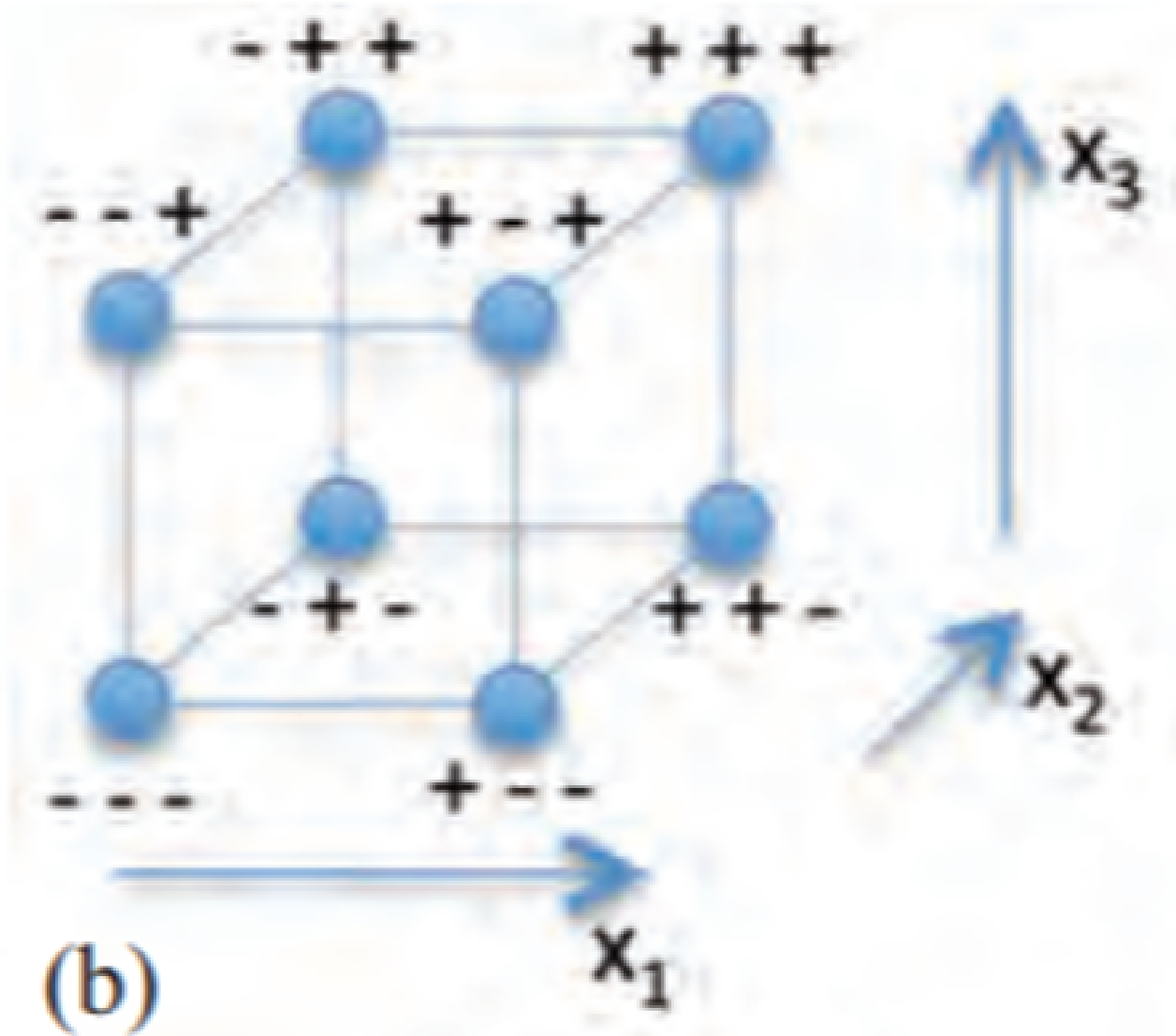
Inc F



Piano sperimentale

Fattori	-	+
T [°C]	28	40
Ibuprofene [mg/l]	6,25	50
DMSO [%v/v]	0	10

# Esperimento	T [°C]	Ibuprofene [mg/l]	DMSO
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-
5	-	+	+
6	-	-	+
7	+	-	+
8	+	+	+



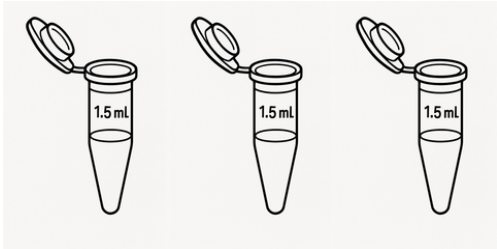
# Esperimento	T [°C]	Ibuprofene [mg/l]	DMSO
9	-	0	+
10	+	0	+

Esperimento

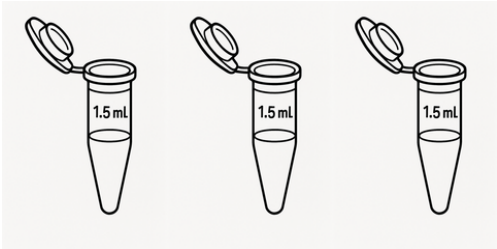
Controllo

Condizione

1

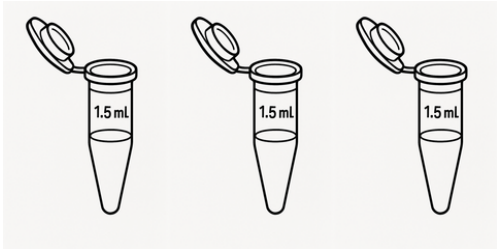


T = 28°C

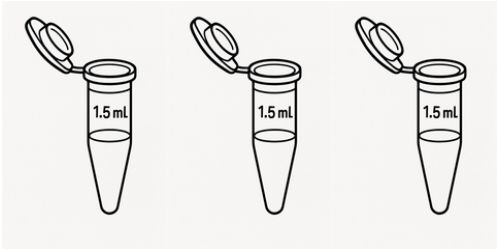


T = 28°C, IBU = 6.25 mg/l

2

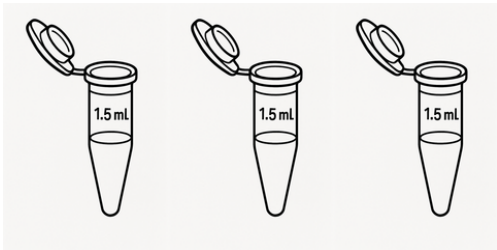


T = 40°C

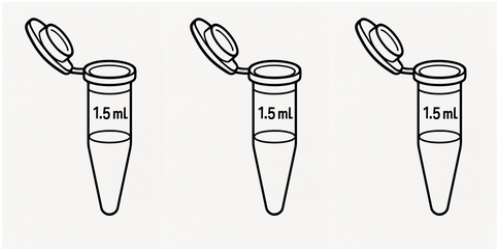


T = 40°C, IBU = 6.25 mg/l

3

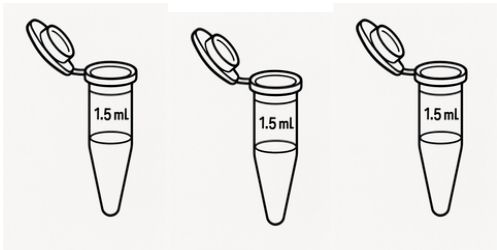


T = 28°C

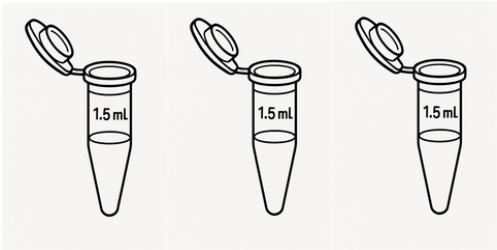


T = 28°C, IBU = 50.0 mg/l

4



T = 40°C



T = 40°C, IBU = 50.0 mg/l

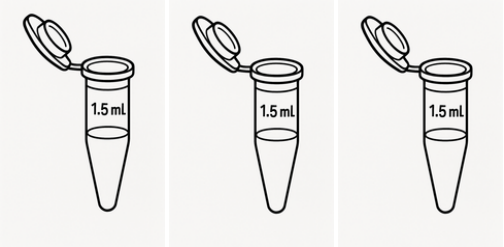
#	T [°C]	Ibuprofen	DMSO
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-

Esperimento

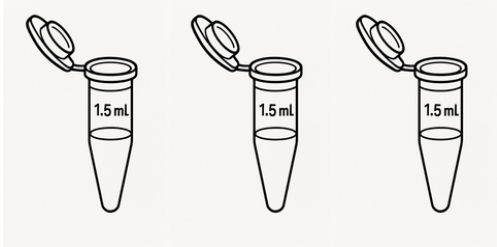
Controllo

Condizione

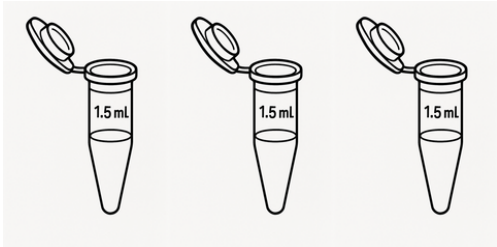
5, 6



T = 28°C, DMSO

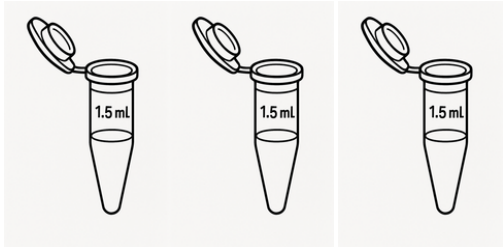


T = 28°C, IBU = 6.25 mg/l, DMSO

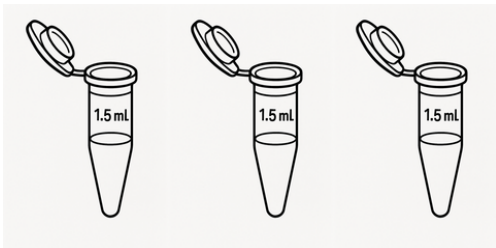


T = 28°C, IBU = 50.0 mg/l, DMSO

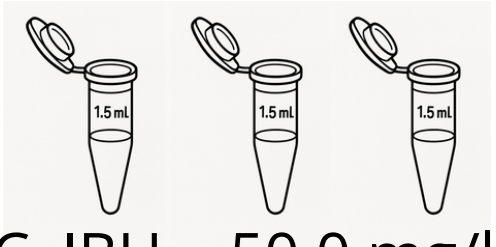
7, 8



T = 40°C, DMSO



T = 40°C, IBU = 6.25 mg/l, DMSO



T = 40°C, IBU = 50.0 mg/l, DMSO

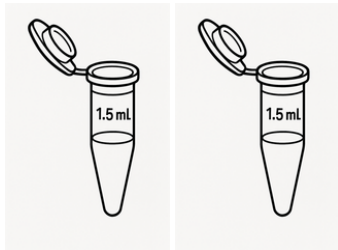
#	T [°C]	Ibuprofen e [mg/l]	DMSO
5	-	+	+
6	-	-	+
7	+	-	+
8	+	+	+

Esperimento

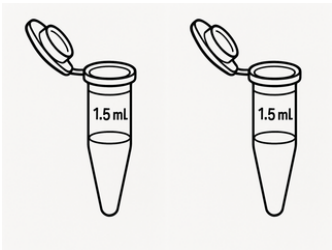
Controllo

Condizione

9



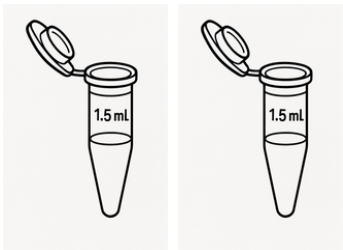
T = 28°C



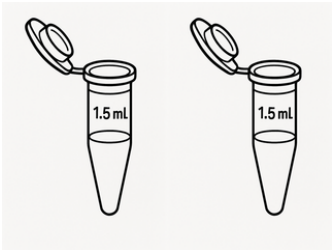
T = 28°C, DMSO

#	T [°C]	IBU [mg/l]	DMSO
9	-	0	+
10	+	0	+

10



T = 40°C



T = 40°C, DMSO

Risultati/ Elaborazione dati

4- Conferma della presenza di ciascun gene: incN (plasmide che ospita il gene blaKPC), incF.

Inc N

“Countable dilution”

[illegible]

TMTC: "To much to count"

- 3- Genotipizzazione per confermare il genotipo del ceppo ricevente.
- 4- Conferma della presenza di ciascun gene: incN (plasmide che ospita il gene blaKPC), incF

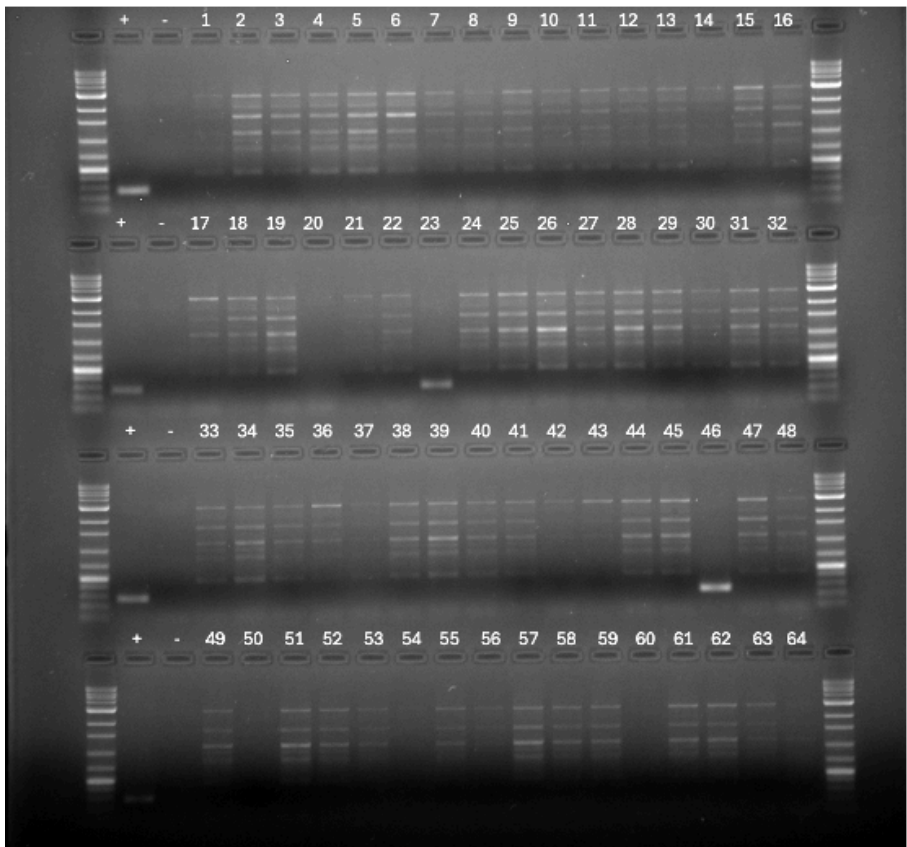
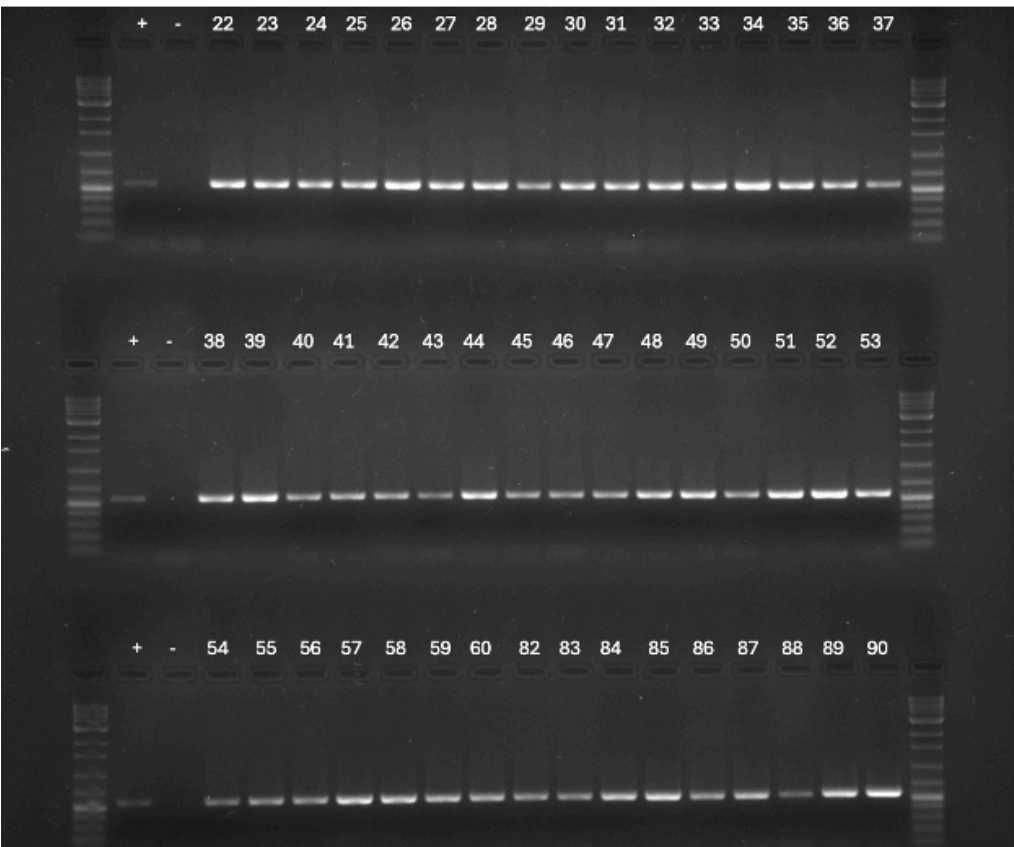
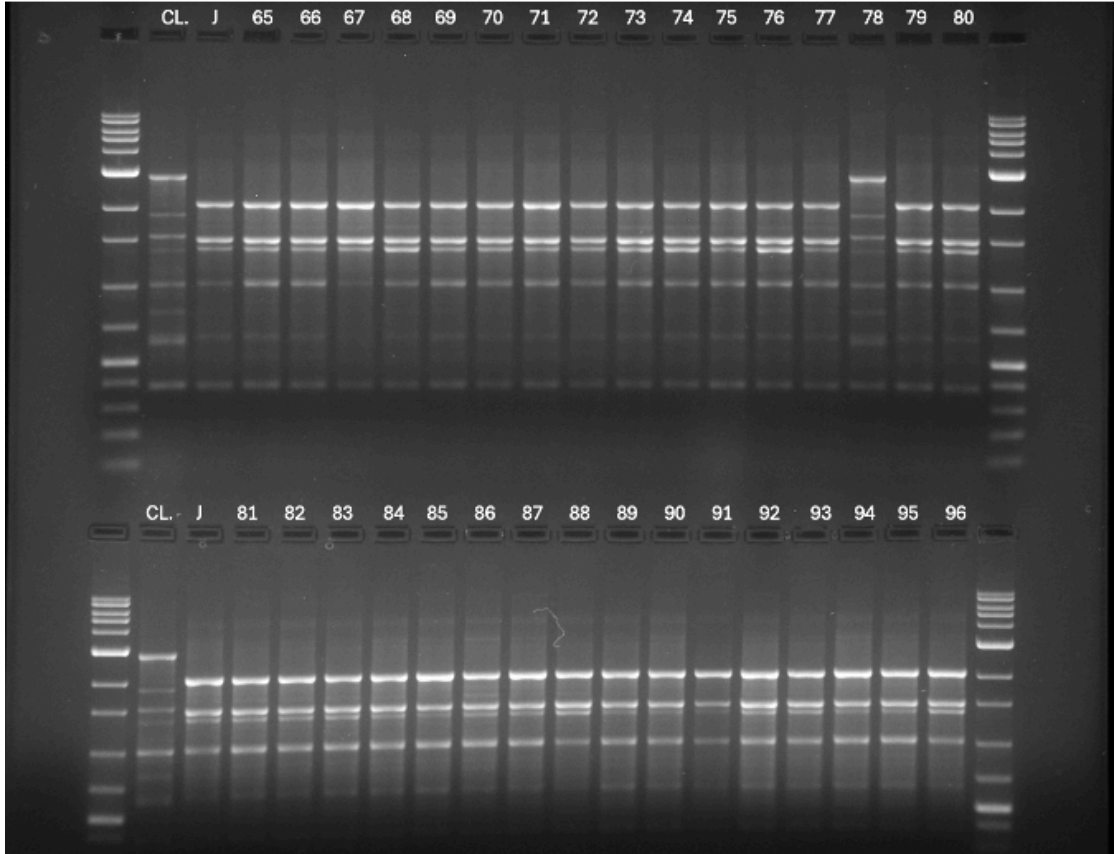
	NO DMSO								DMSO								DMSO vs NO DMSO			
	T = 28°C, 18h = 8.25 mg/l		T = 40°C, 18h = 8.25 mg/l		T = 28°C, 18h = 50 mg/l		T = 40°C, 18h = 50 mg/l		T = 28°C, 18h = 8.25 mg/l		T = 40°C, 18h = 8.25 mg/l		T = 28°C, 18h = 50 mg/l		T = 40°C, 18h = 50 mg/l		T = 28°C		T = 40°C	
	Control	8hU	Control	8hU	Control	8hU	Control	8hU	DMSO Control	8hU	DMSO Control	8hU	DMSO Control	8hU	DMSO Control	8hU	Control	DMSO	Control	DMSO
Number of samples	35	29	40	51	51	42	45	54	27	25	27	27	27	21	27	30	24	24	24	24
Positive to RAPD (.)	35	28	40	50	51	38	44	53	27	25	27	27	27	21	27	30	24	24	24	24
Positive to RAPD (%)	100,0%	96,6%	100,0%	98,0%	100,0%	90,5%	97,8%	98,1%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%	100,0%
Positive to inc N	35	28	40	50	51	37	44	53	27	25	27	27	27	21	27	30	24	24	24	24
True transconjugants	35	28	40	50	51	37	44	53	27	25	27	27	27	21	27	30	24	24	24	24
True transconjugants (%)	100,0%	96,6%	100,0%	98,0%	100,0%	88,1%	97,8%	98,1%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Overall Positive to inc F	0	1	2	1	0	4	2	1	0	0	0	0	0	0	0	0				
True transconjugants positive to inc F	0	0	2	0	0	0	1	0	0	0	0	0	0	0	0	0				
True transconjugants positive to inc F (%)	0,0%	0,0%	5,0%	0,0%	0,0%	0,0%	2,2%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%				

Veri transcongiuganti

RAPD

Inc N

Inc F



TRUE CFU/100ul

$$\text{True CFU/ml} = (\text{True CFU}_{100\text{ul}} \times \text{Dilution Factor} \times 1000) / 100$$

TRUE CFU/ml

TRUE log10CFU/ml

Frequenza di coniugazione

$$(\text{True CFU/ml}) / (\# \text{donatori})$$

Obiettivo valori logaritmici: osservare differenze sostanziali dal punto di vista microbiologico (soglia > **0.5 log**)

True CFU_100ul

			NO DMSO								DMSO								DMSO vs NO DMSO			
			T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C		T = 40°C	
			Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO
Condition Replicate	A	-1	THTC	THTC	157	200	165	54	252	211	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC	114	186	229	301
			THTC	THTC	226	214	195	93	242	233	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC	122	157	192	309
			THTC	THTC	182,9	152,6	166	98	198	213	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC	127	148	233	315
		-2	27	36	4	5	16	8	15	12	10	10	10	36	10	12	10	13	9	1	3	1
			29	33	7	2	13	10	11	5	14	7	8	33	14	6	8	11	13	4	3	3
			37	30	11	3	14	7	12	3	12	7	7	29	12	10	7	18	8	5	5	3
	B	-1	3	4	0	1	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
			7	6	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
			3	18	0	0	1	0	2	0	1	1	1	0	0	0	1	5	0	0	0	0
		-2	THTC	THTC	201	179	170	165	142	196	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC	86	108	202	306
			THTC	THTC	139	140	145	200	177	177,5	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC	75	81	238	297
			THTC	THTC	106	160	159	163	128	271	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC	71	129	178	294
	C	-1	26,25	40	10	7	13	23	9	5	7	8	30	21	7	9	30	47	1	3	3	2
			60	26	7	9	11	18	10	11	15	8	19	35	15	5	19	18	5	5	3	7
			49	35	6	8	10	17	14	14	7	10	35	16	7	8	35	37	4	2	4	8
		-2	4	16	0	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	1
			3	5	3	3	1	2	0	1	1	1	0	0	0	0	0	3	0	0	0	0
			2	6	0	2	2	2	0	1	1	1	0	0	0	0	0	0	0	0	0	0
		-1	THTC	THTC	221	131	128	129	137,8	157	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC				
			THTC	THTC	263	157	168	90	170	142	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC				
			THTC	THTC	254	141	132	122	144	233	THTC	THTC	THTC	THTC	THTC	THTC	THTC	THTC				
		-2	26	21,3	15	18	3	16	4	3	11	15	18	31	11	14	18	24				
			33	16	9	15	4	8	3	5	13	12	20	27	13	12	20	23				
			41	26	11	11	5	8	5	8	12	10	26	25	12	15	16	28				
		-3	8	3	2	1	0	1	0	0	0	1	0	0	0	0	0	2				
			5	3	0	0	0	1	0	0	0	1	0	1	0	0	1	1				
			25	6	1	1	1	0	0	0	1	0	0	0	0	0	0	0				

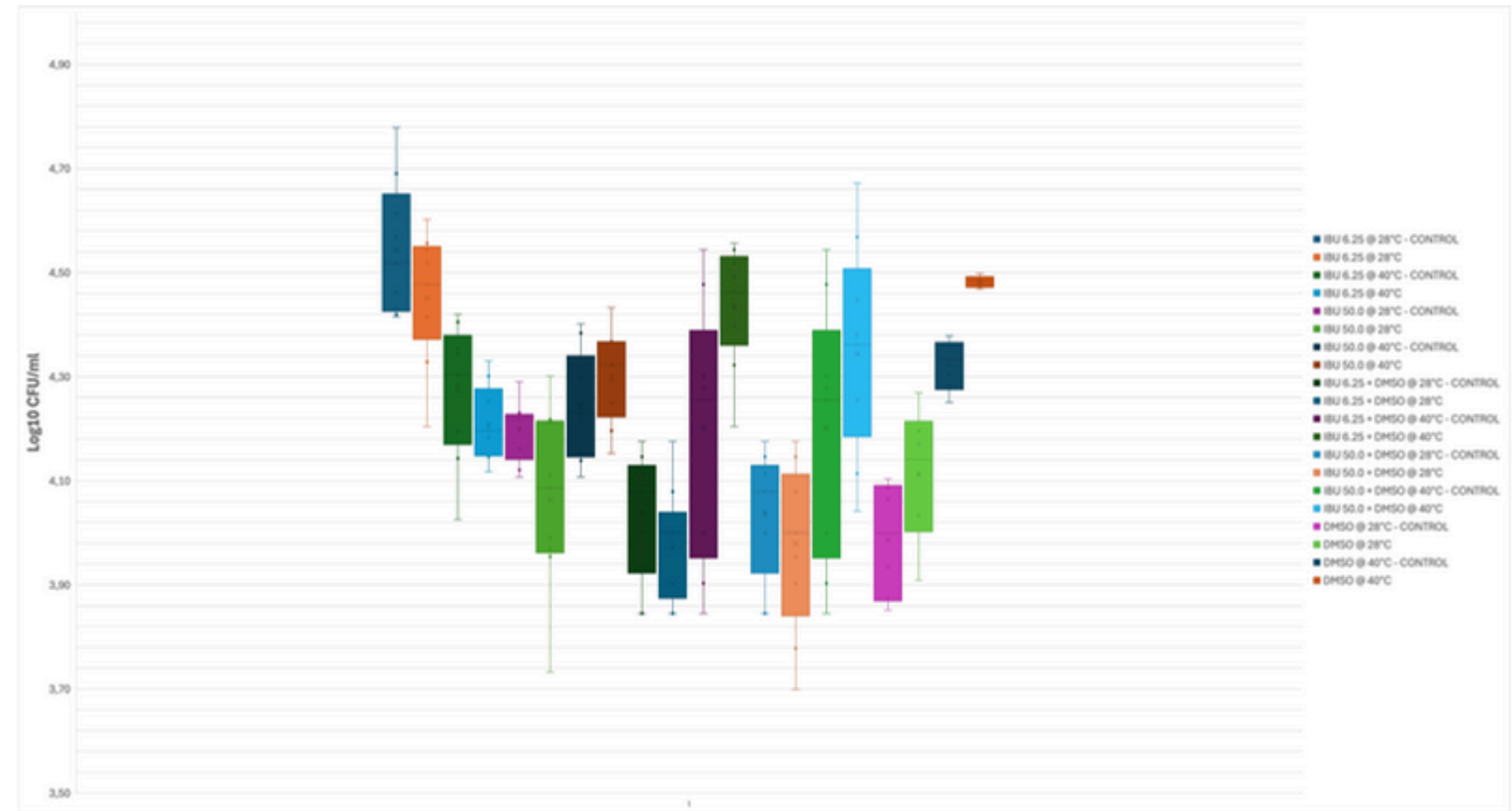
True CFU/ml = (True CFU_100ul x Dilution Factor x 1000)/100

True CFU/ml - "Countable dilution"

			NO DMSO								DMSO								DMSO vs NO DMSO																							
			T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C		T = 40°C																					
			Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO																				
A	27000		36000		15700		20000		16500		5400		25200		21100		10000		10000		10000		12000		10000		13000		11600		18600		22900		30100							
	29000		33000		22600		21400		19500		9300		24200		23300		14000		7000		8000		33000		14000		6000		8000		11000		12200		15700		19200		30900			
	37000		30000		19290		15260		16600		9800		19800		21000		12000		7000		7000		29000		12000		10000		7000		18000		12700		14800		23000		31500			
B	26250		40000		20100		17900		17000		16500		14200		19600		7000		8000		30000		21000		7000		9000		30000		47000		8600		10800		20200		30600			
	60000		26000		13900		14000		14500		20000		17700		17750		15000		8000		19000		35000		15000		5000		19000		18000		7500		8100		23800		29700			
	49000		35000		10600		16000		15900		16300		12800		27100		7000		10000		35000		16000		7000		8000		35000		37000		7100		12900		17800		29400			
C	26000		21300		22100		13100		12800		12900		13760		15700		11000		15000		18000		31000		11000		14000		18000		24000											
	33000		16000		26300		15700		16800		9000		17000		14200		13000		12000		20000		27000		13000		12000		20000		23000											
	41000		26000		25400		14100		13200		12200		14400		23300		12000		10000		16000		21000		12000		15000		16000		28000											
Mean value			36472		29256		19554		16384		15867		12378		17896		20339		11222		9667		18111		28111		11222		10111		18111		24333		9950		13483		21167		30367	
Standard deviation			11742		7654		6277		2831		2082		4569		4423		4042		2819		2598		9532		6623		2819		3444		9532		11597		2502		3724		2448		784	

Log10CFU/ml

	NO DMSO								DMSO								DMSO vs NO DMSO			
	T=28°C, IBU= 6,25 mg/l		T= 40°C, IBU= 6,25 mg/l		T= 28°C, IBU= 50 mg/l		T=40°C, IBU= 50 mg/l		T=28°C, IBU= 6,25 mg/l		T= 40°C, IBU= 6,25 mg/l		T= 28°C, IBU= 50 mg/l		T=40°C, IBU= 50 mg/l		T=28°C		T= 40°C	
	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO
A	4,43	4,56	4,20	4,30	4,22	3,73	4,40	4,32	4,00	4,00	4,00	4,56	4,00	4,08	4,00	4,11	4,06	4,27	4,36	4,48
	4,46	4,52	4,35	4,33	4,29	3,97	4,38	4,37	4,15	3,85	3,90	4,52	4,15	3,78	3,90	4,04	4,09	4,20	4,28	4,49
	4,57	4,48	4,29	4,18	4,22	3,99	4,30	4,32	4,08	3,85	3,85	4,46	4,08	4,00	3,85	4,26	4,10	4,17	4,36	4,50
B	4,42	4,60	4,30	4,25	4,23	4,22	4,15	4,29	3,85	3,90	4,48	4,32	3,85	3,95	4,48	4,67	3,93	4,03	4,31	4,49
	4,78	4,41	4,14	4,15	4,16	4,30	4,25	4,25	4,18	3,90	4,28	4,54	4,18	3,70	4,28	4,26	3,88	3,91	4,38	4,47
	4,69	4,54	4,03	4,20	4,20	4,21	4,11	4,43	3,85	4,00	4,54	4,20	3,85	3,90	4,54	4,57	3,85	4,11	4,25	4,47
C	4,41	4,33	4,34	4,12	4,11	4,11	4,14	4,20	4,04	4,18	4,26	4,49	4,04	4,15	4,26	4,38				
	4,52	4,20	4,42	4,20	4,23	3,95	4,23	4,15	4,11	4,08	4,30	4,43	4,11	4,08	4,30	4,36				
	4,61	4,41	4,40	4,15	4,12	4,09	4,21	4,37	4,08	4,00	4,20	4,40	4,08	4,18	4,20	4,45				
Mean value	4,54	4,45	4,28	4,21	4,29	4,06	4,24	4,30	4,04	3,97	4,20	4,44	4,04	3,98	4,20	4,34	3,99	4,11	4,32	4,48
Standard deviation	0,13	0,13	0,13	0,07	0,06	0,17	0,10	0,09	0,12	0,11	0,24	0,11	0,12	0,16	0,24	0,20	0,11	0,13	0,05	0,01
Median	4,52	4,48	4,30	4,20	4,22	4,09	4,23	4,32	4,08	4,00	4,26	4,46	4,08	4,00	4,26	4,36	4,00	4,14	4,33	4,48
Δlog	-0,09		-0,07		-0,13		0,06		-0,06		0,24		-0,06		0,14		0,13		0,16	

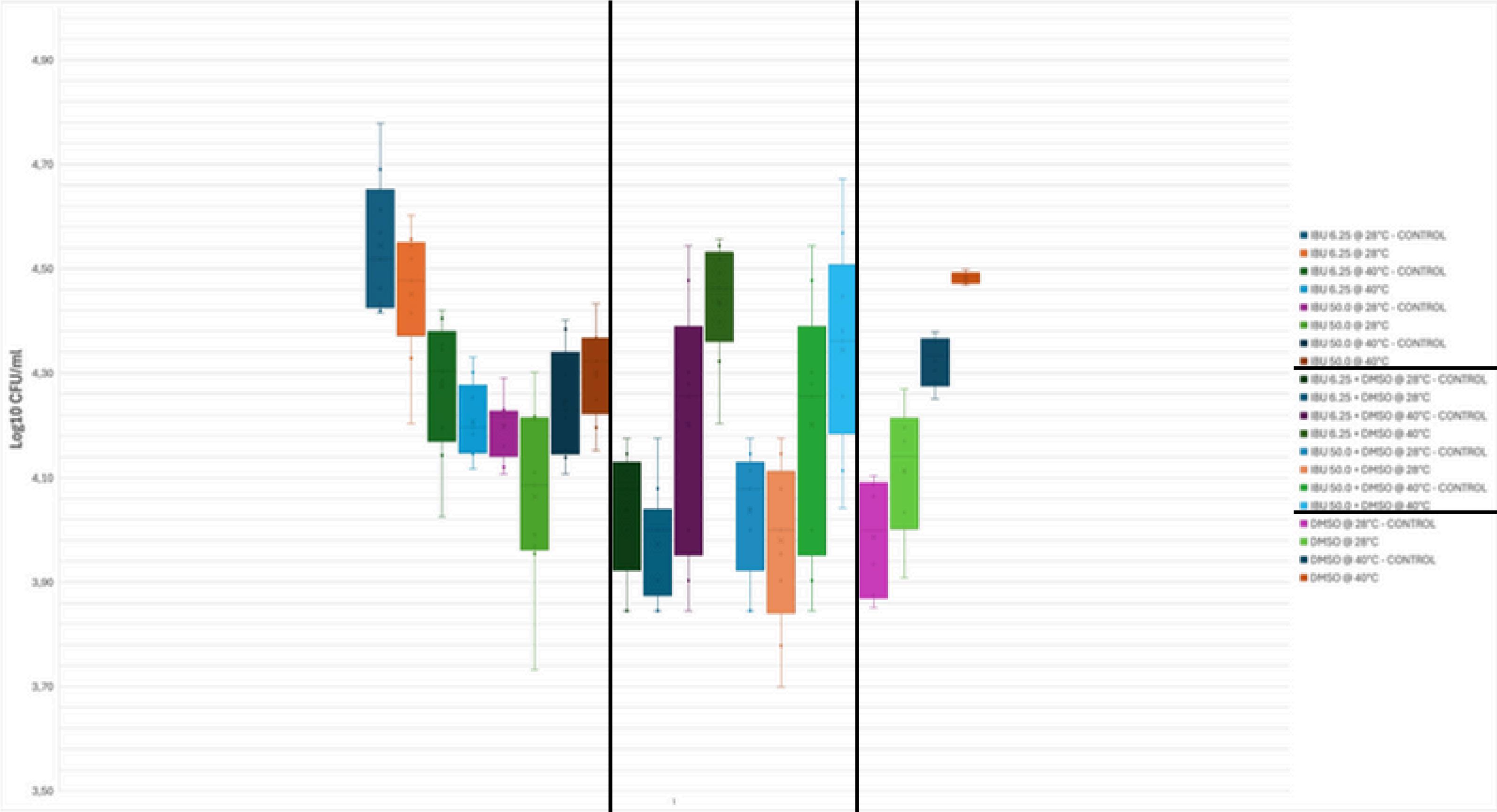


ESPERIMENTO

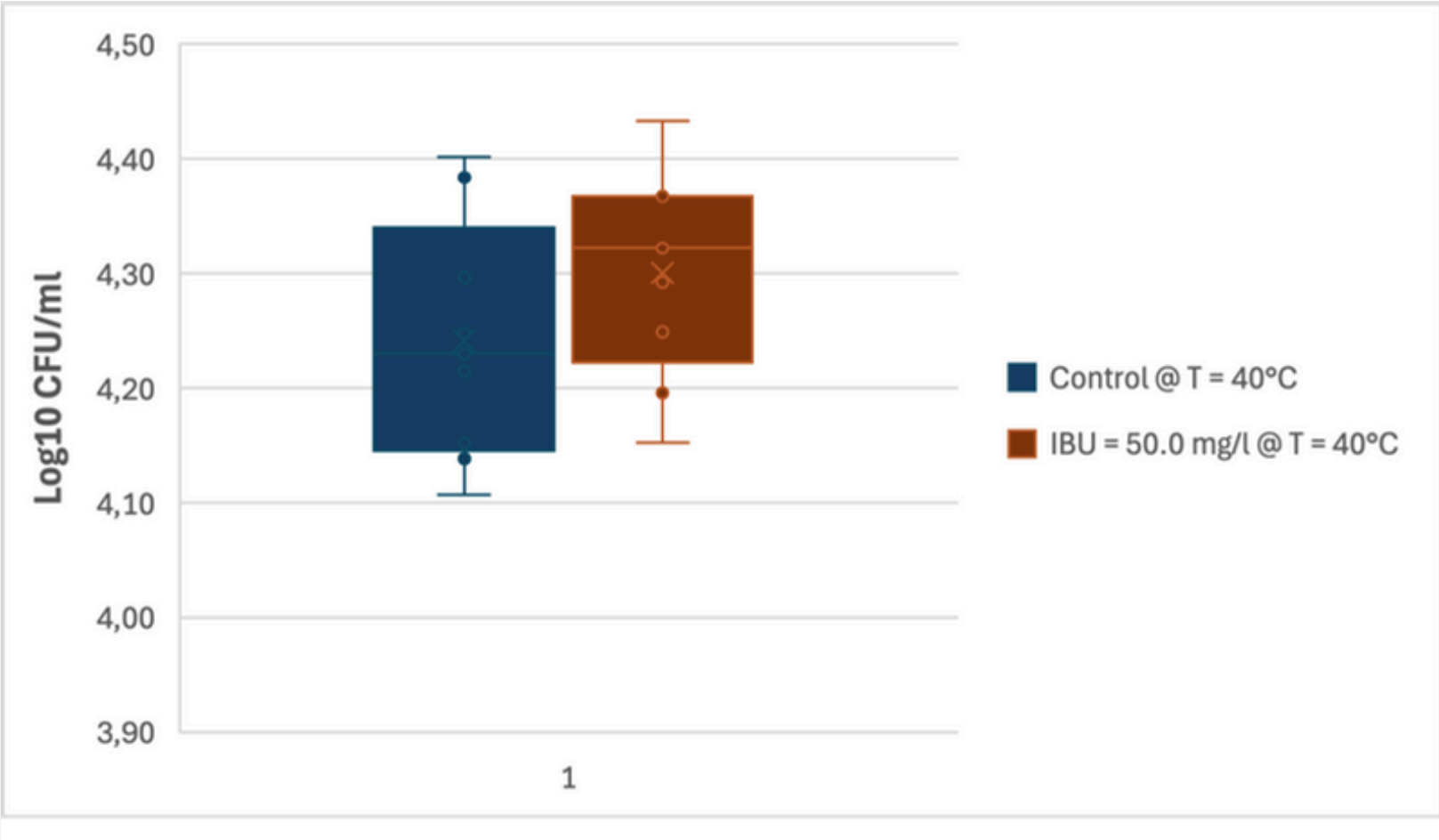
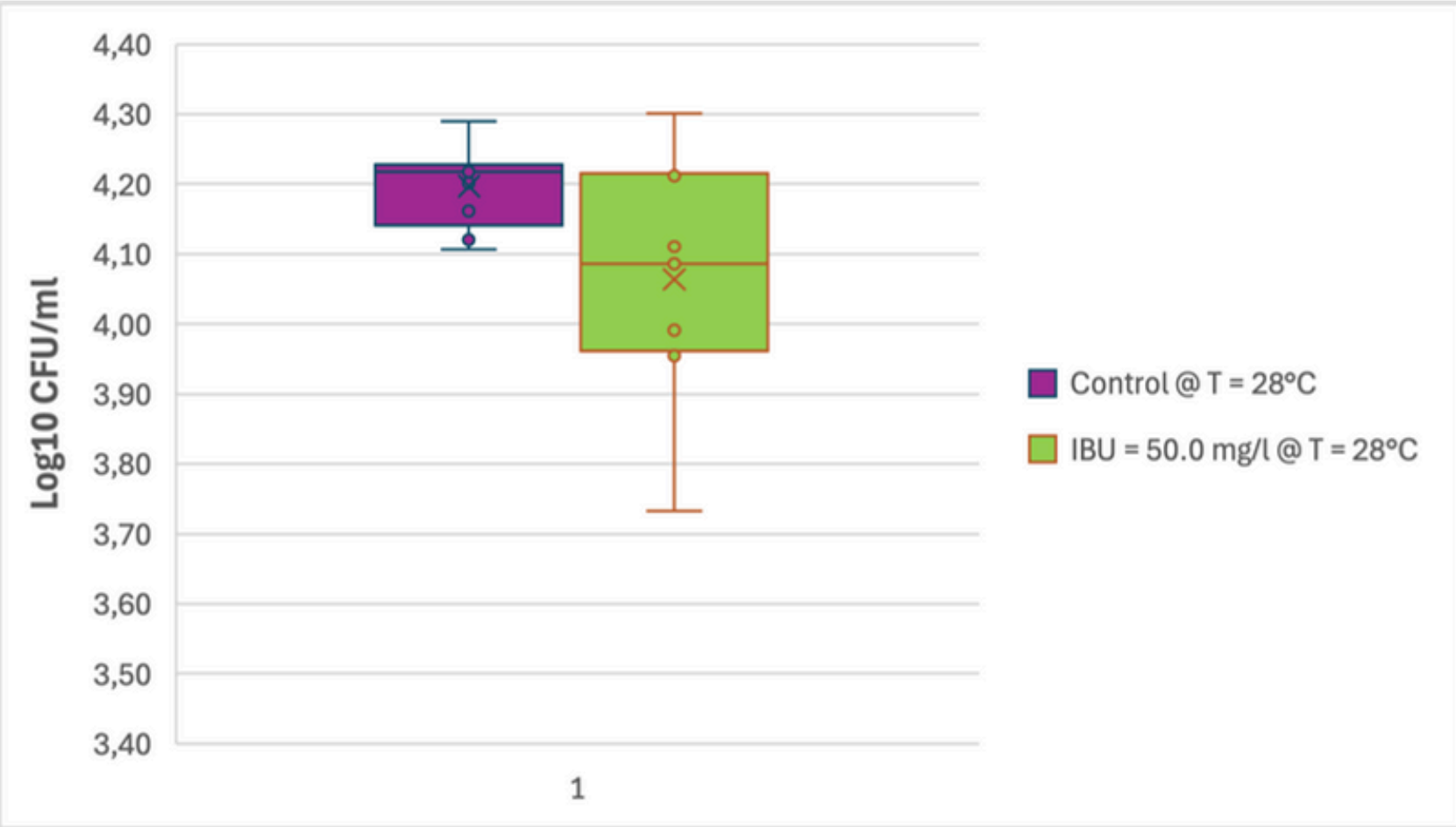
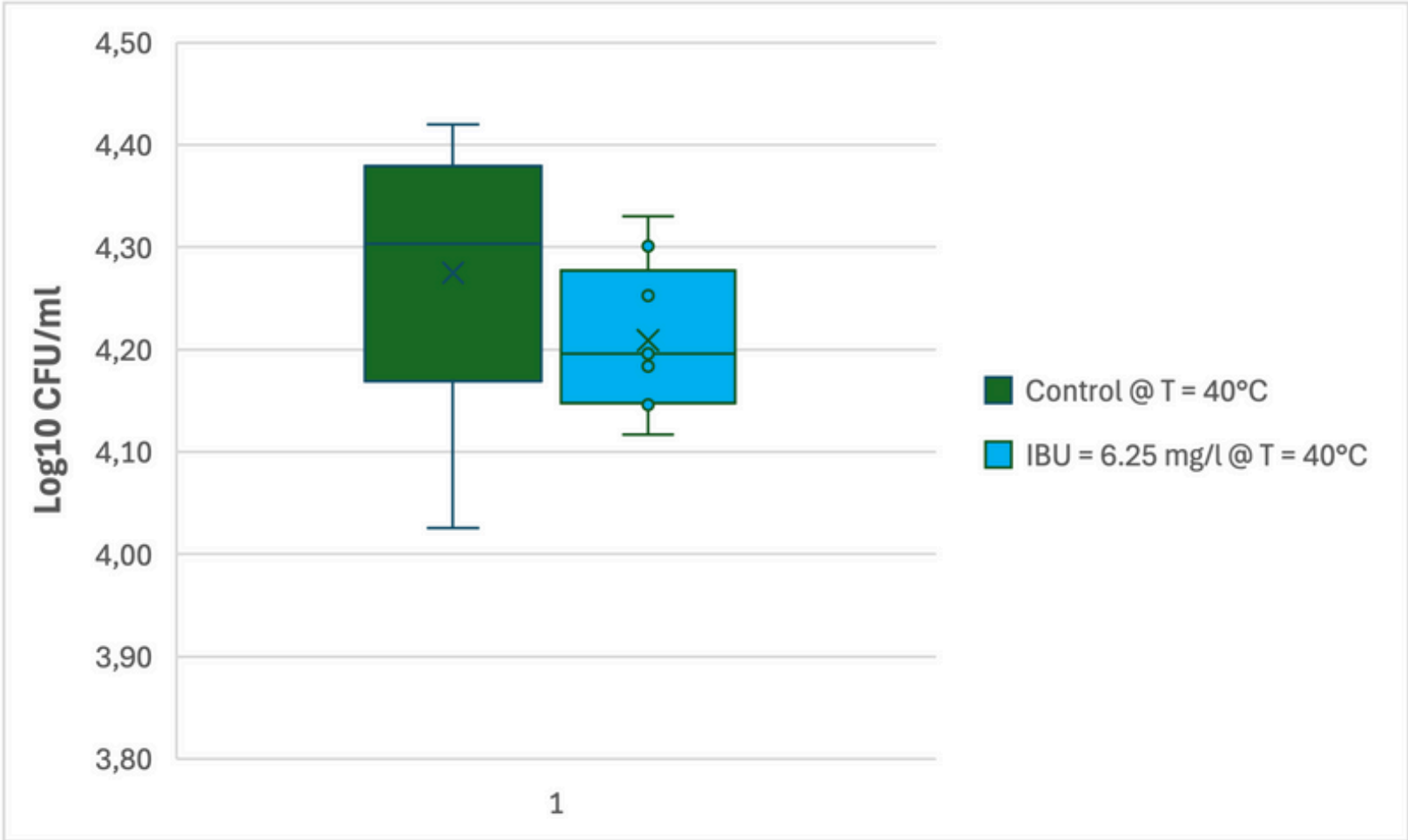
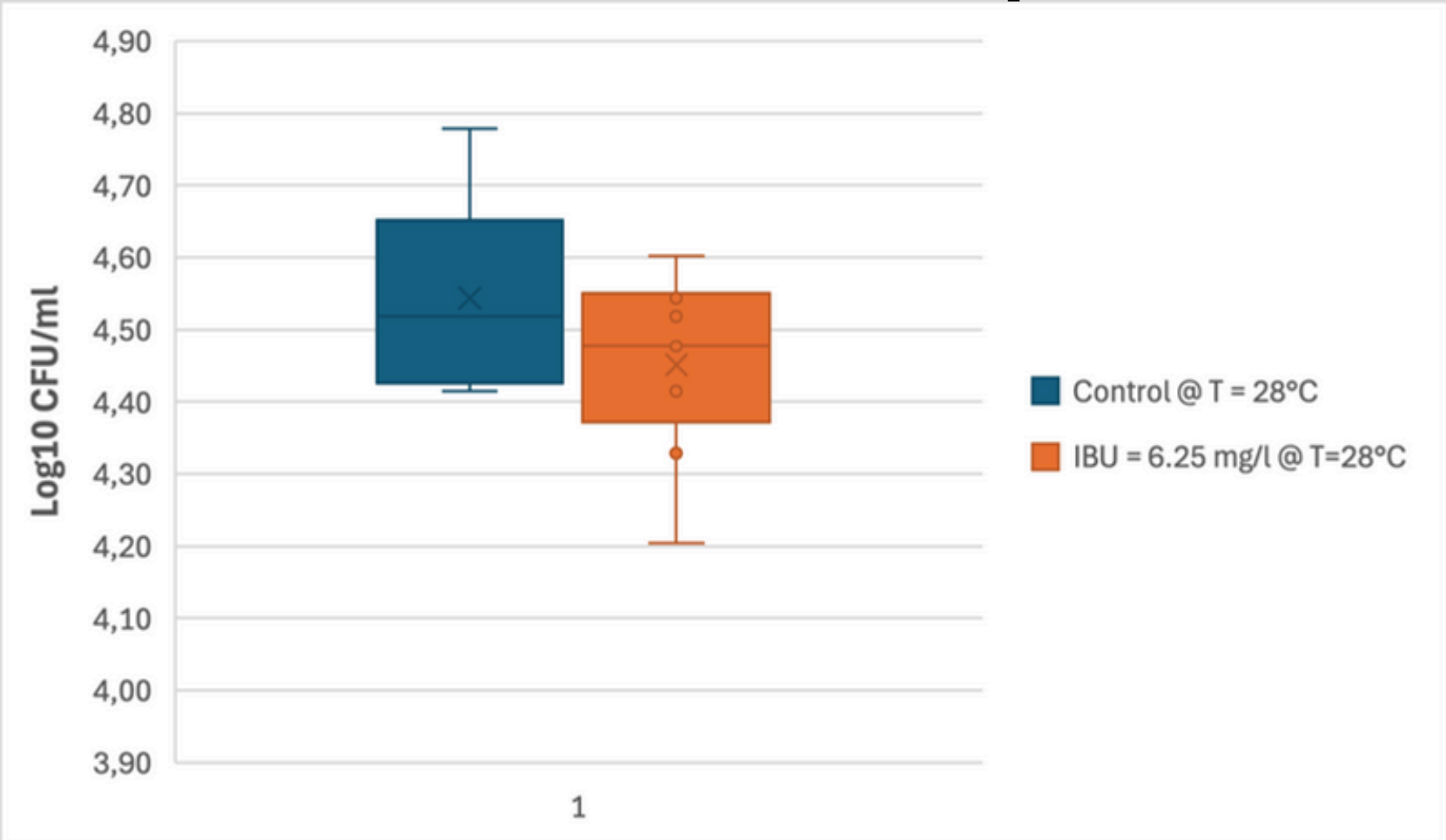
1-4

5-8

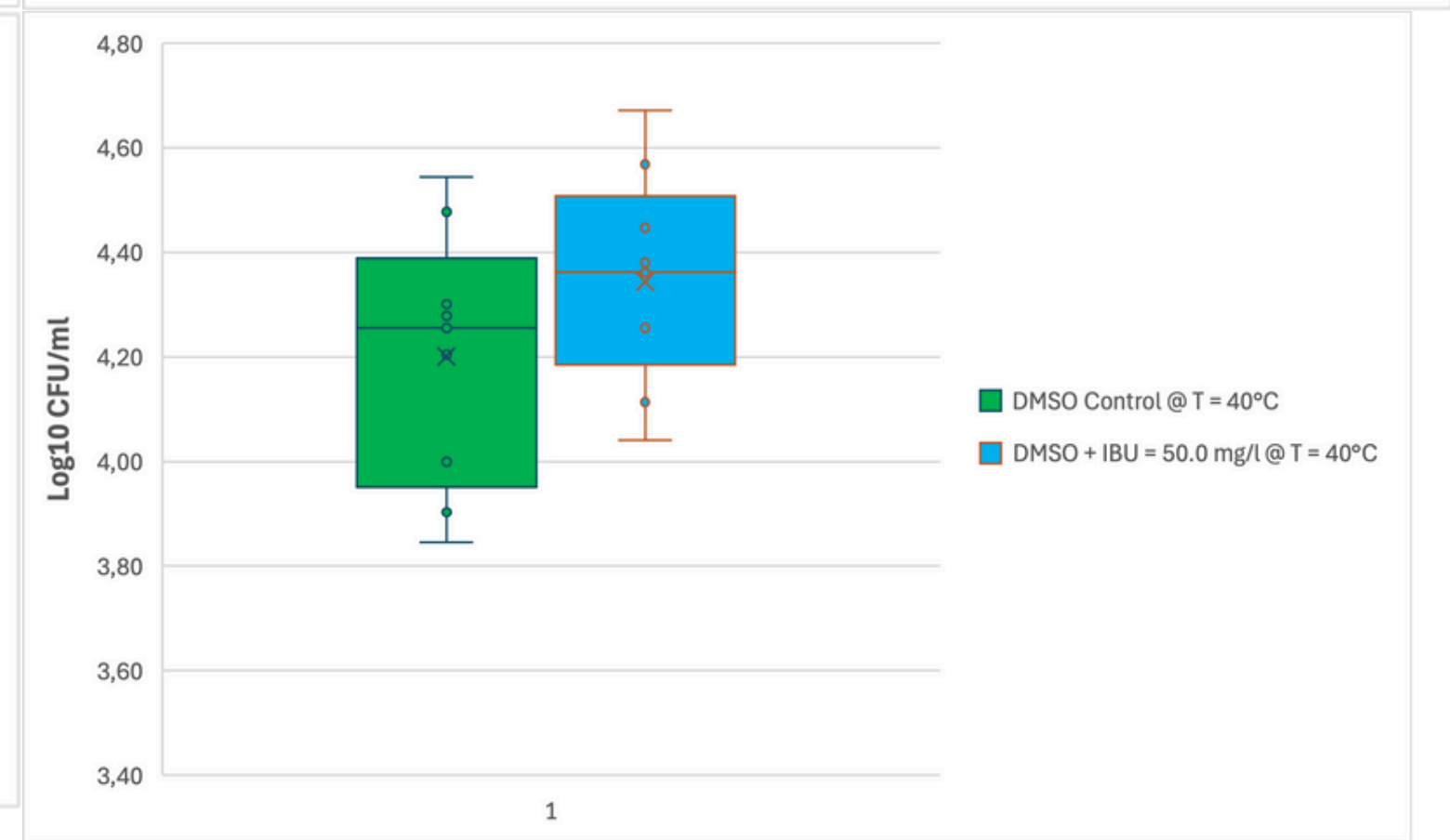
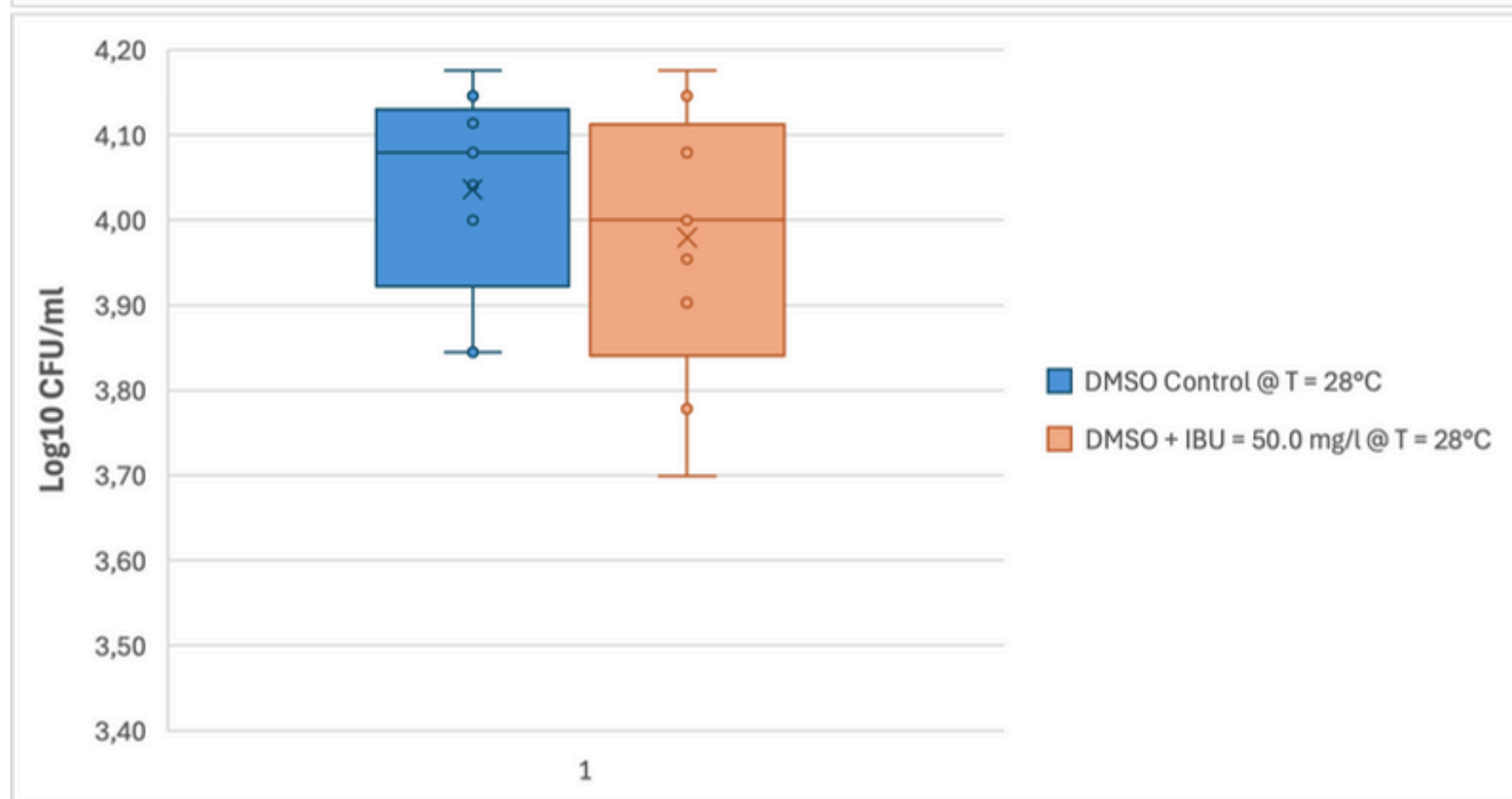
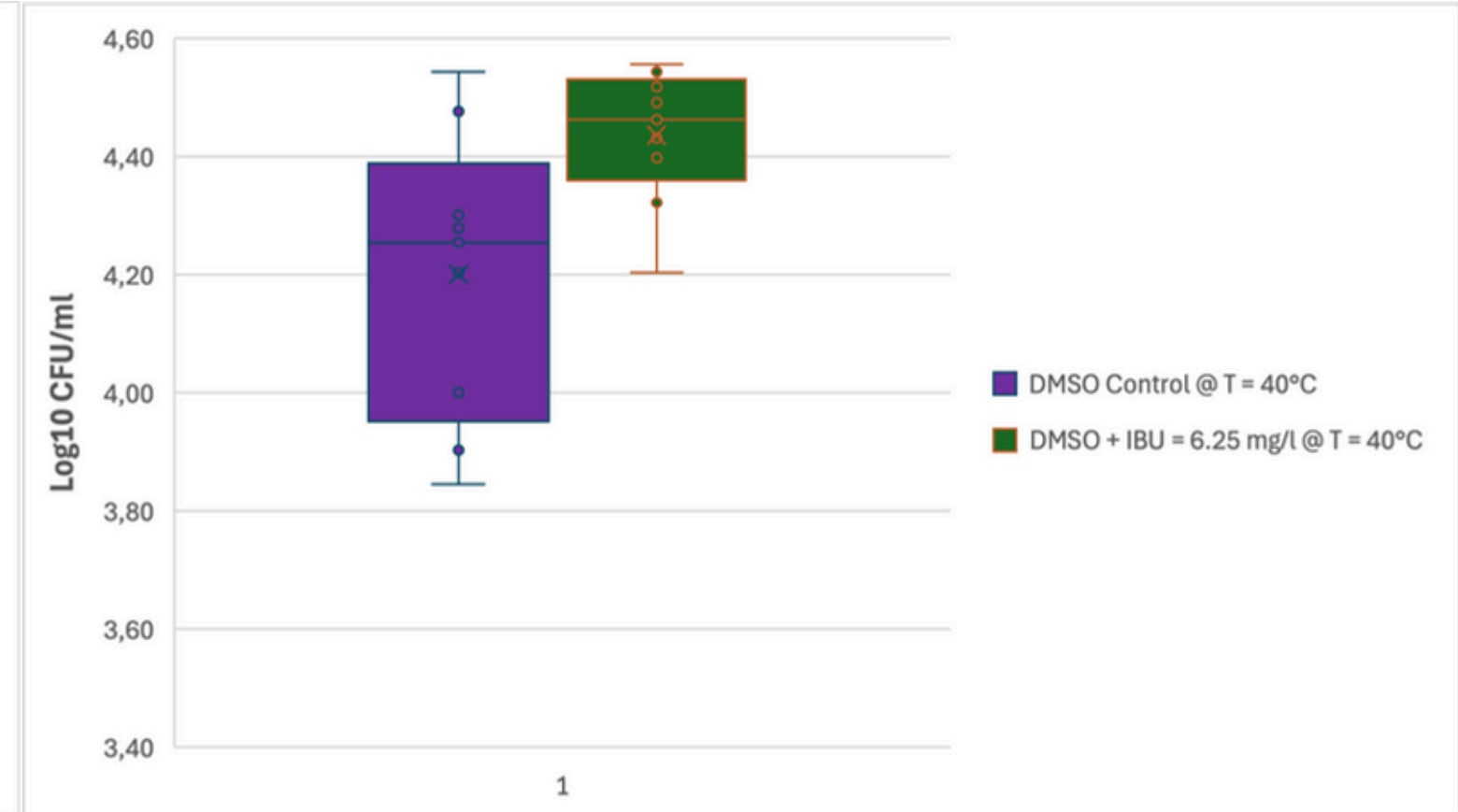
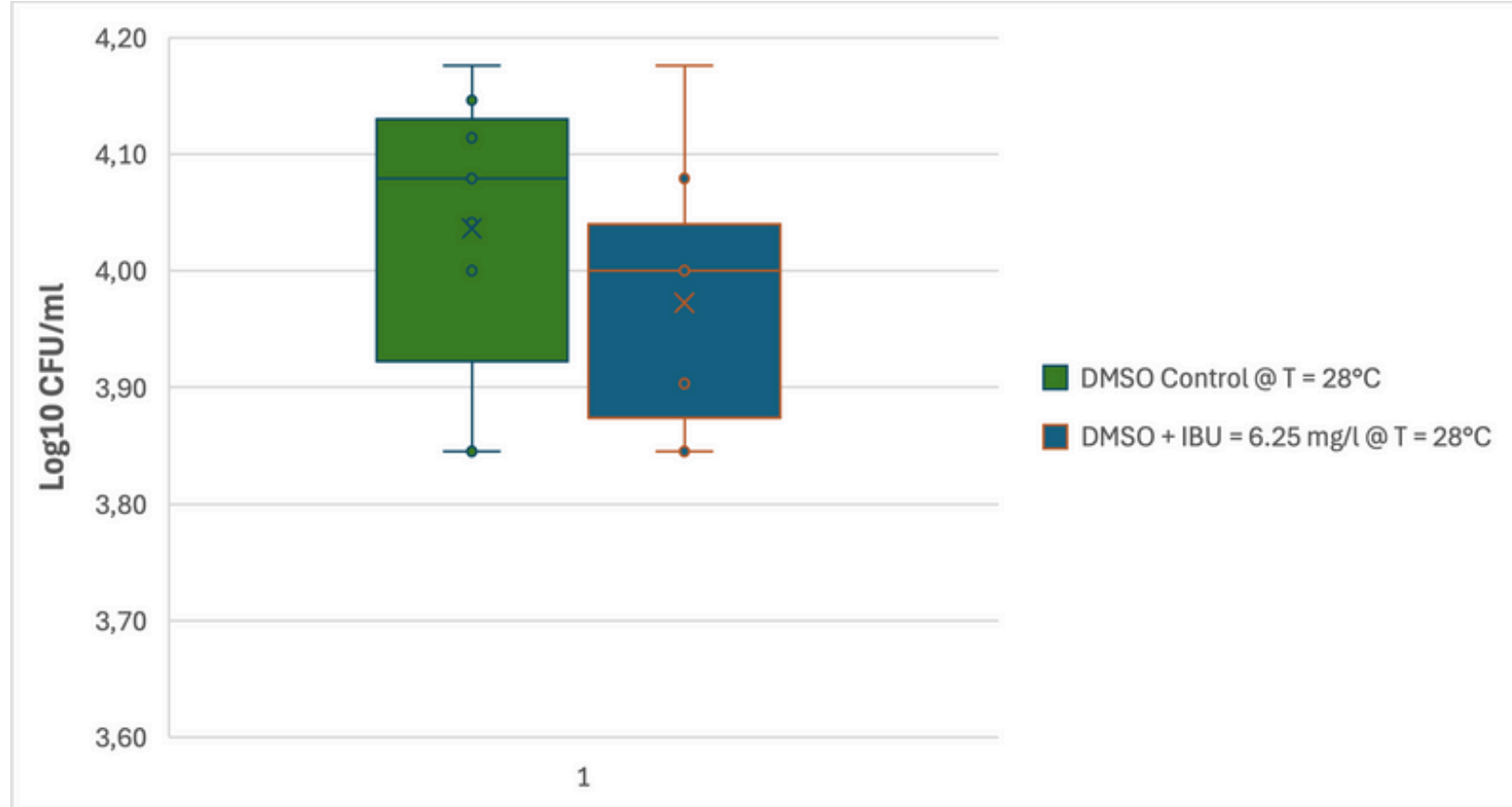
9-10



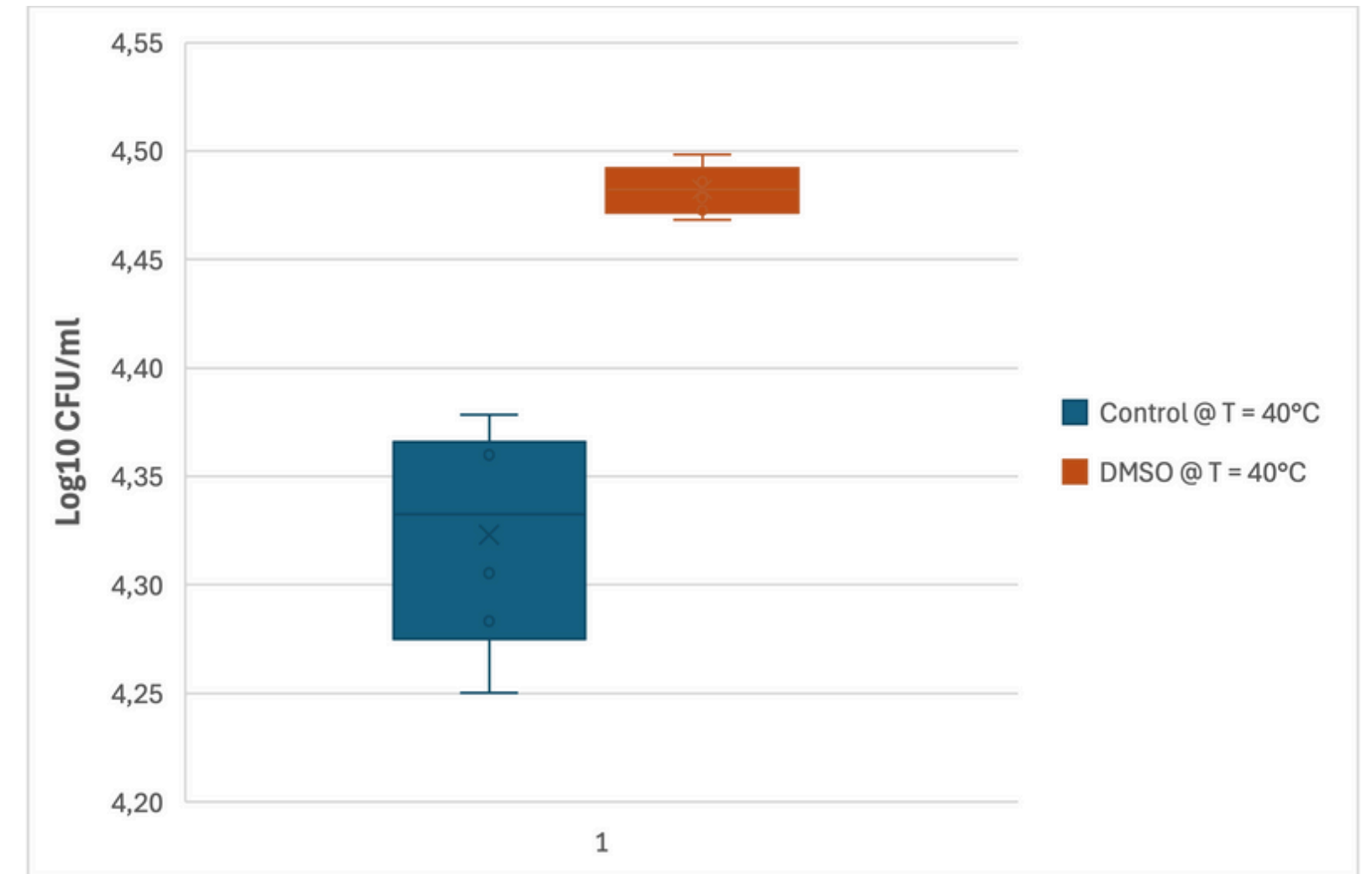
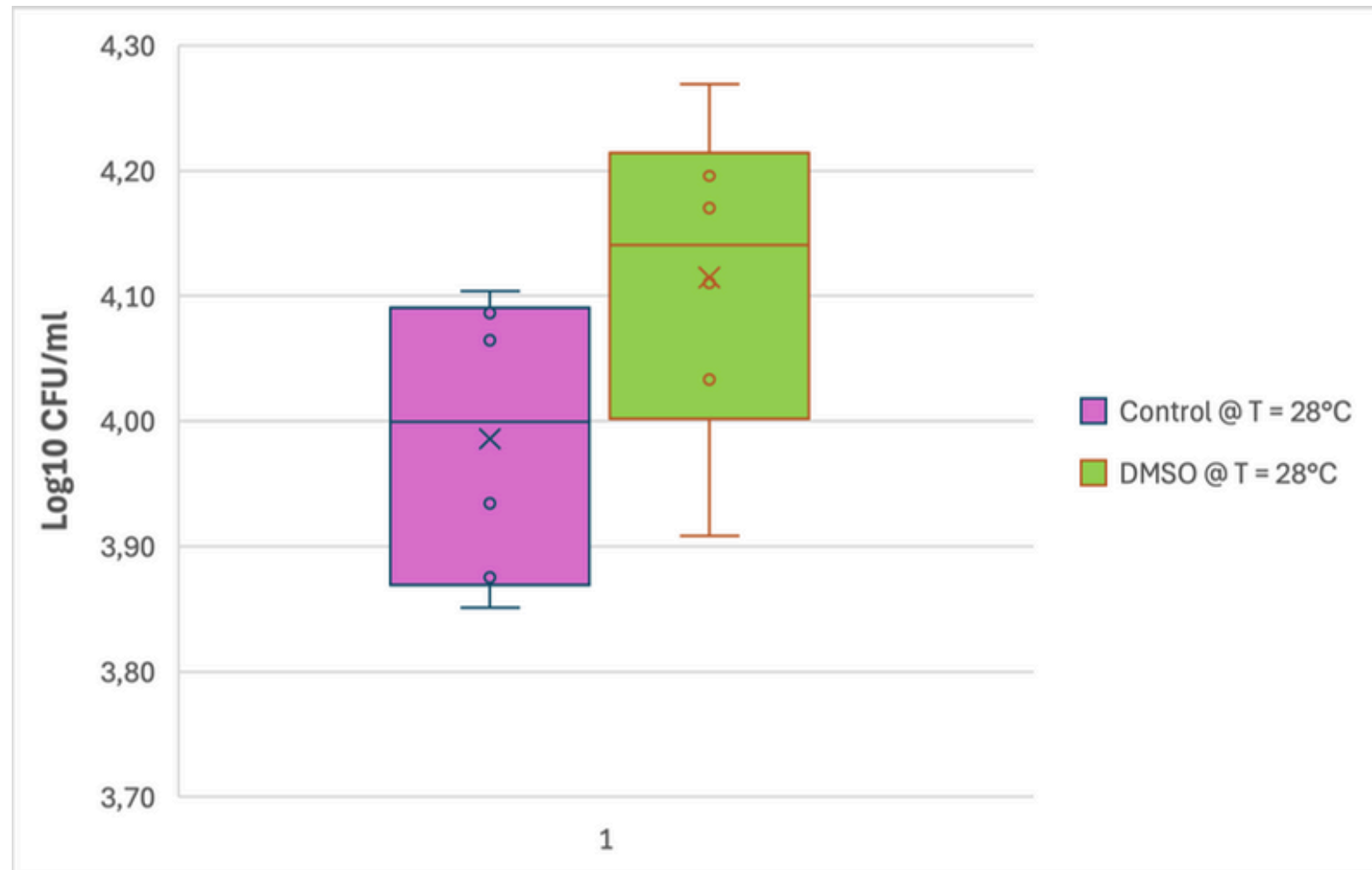
Esperimenti 1-4 (NO DMSO)

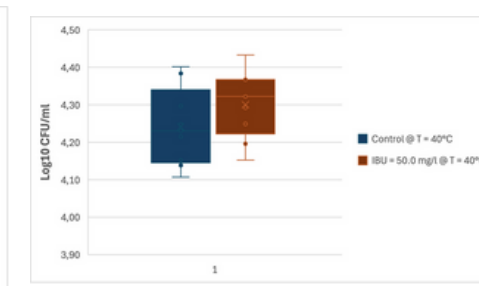
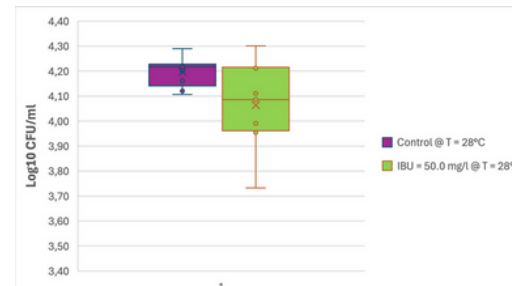
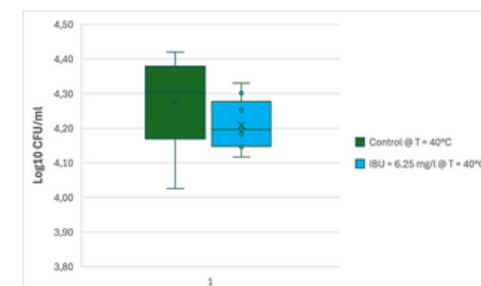
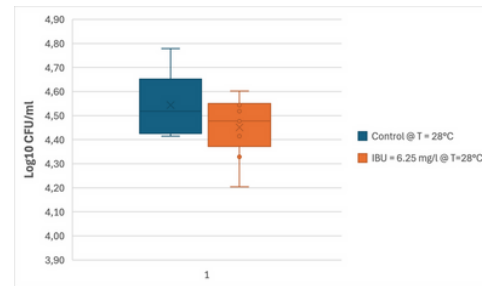


ESPERIMENTI 5-8 (DMSO)

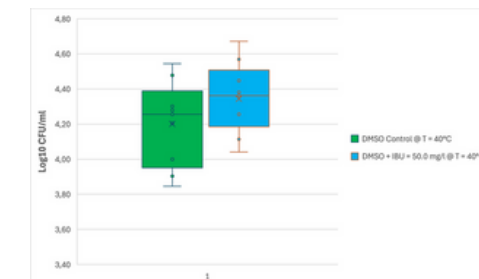
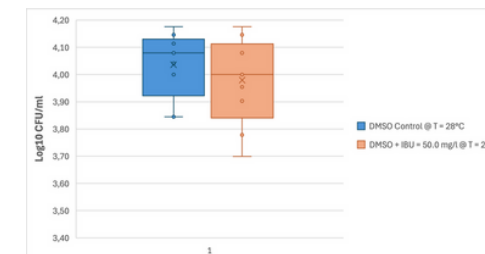
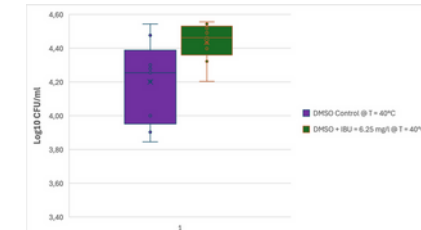
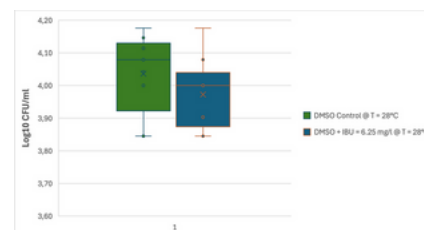


ESPERIMENTI 9-10 (DMSO VS NO DMSO)





	Esperimento 1		Esperimento 2		Esperimento 3		Esperimento 4	
	T = 28°C, IBU= 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l	
	Control	Condition	Control	Condition	Control	Condition	Control	Condition
	4,54	4,45	4,28	4,21	4,20	4,06	4,24	4,30
Δlog mean values	-0,09		-0,07		-0,13		0,06	
Absolute variation	1,24		1,16		1,36		1,15	



	Esperimento 5		Esperimento 6		Esperimento 7		Esperimento 8	
	T = 28°C, IBU= 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l	
	Control	IBU	Control	IBU	Control	IBU	Control	IBU
	4,04	3,97	4,20	4,44	4,04	3,98	4,20	4,34
Δlog mean values	-0,06		0,24		-0,06		0,14	
Absolute variation	1,16		1,72		1,14		1,39	

TRUE CFU/100ul

$$\text{True CFU/ml} = (\text{True CFU}_{100\text{ul}} \times \text{Dilution Factor} \times 1000) / 100$$

TRUE CFU/ml

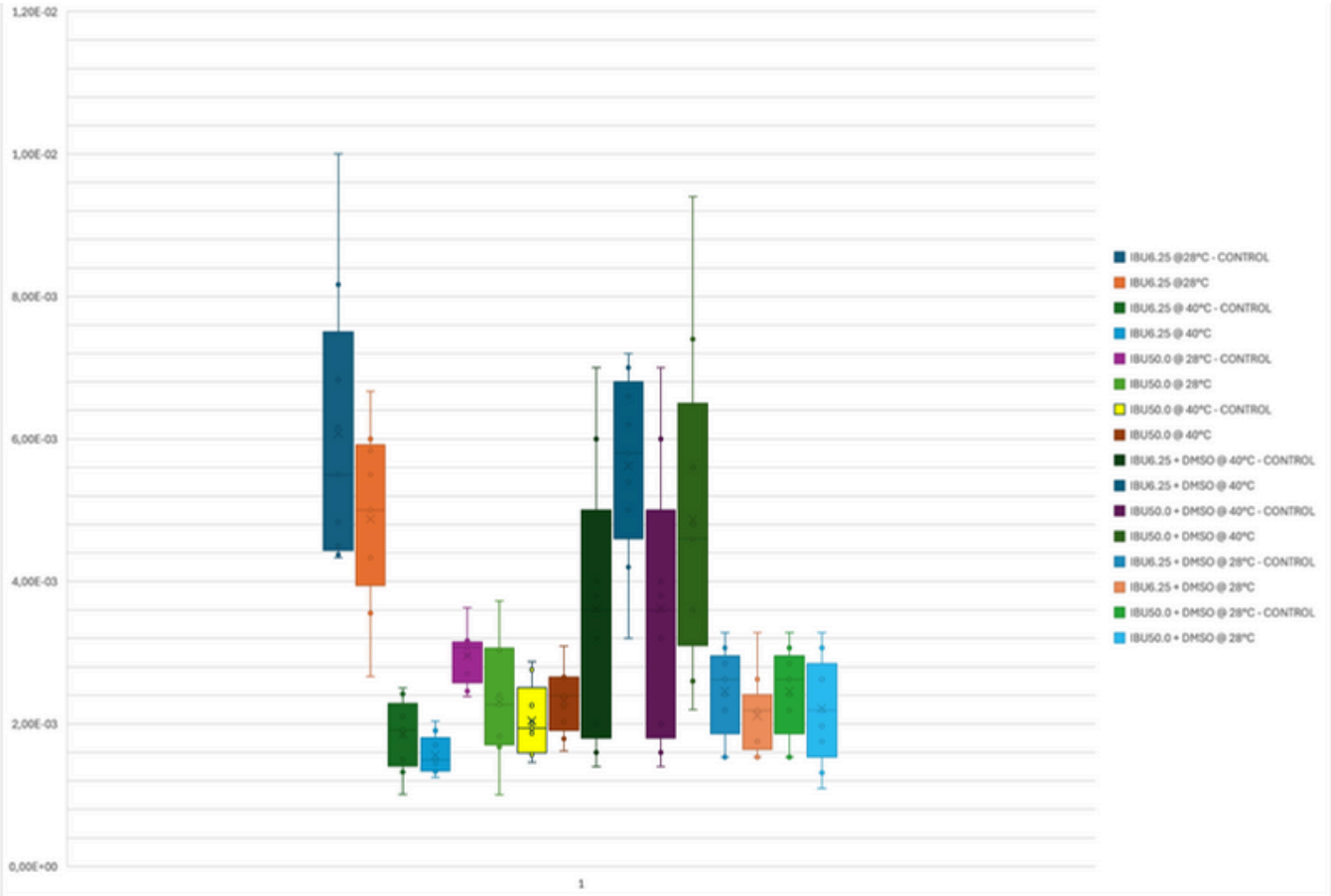
TRUE log₁₀CFU/ml

Obiettivo valori logaritmici: osservare differenze sostanziali dal punto di vista microbiologico (soglia > **0.5 log**)

Frequenza di coniugazione
(True CFU/ml) / (#donatori)

Frequenza di Coniugazione

NO DMSO								
	T = 28°C, IBU= 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l	
	Control	IBU	Control	IBU	Control	IBU	Control	IBU
A	4,50E-03	6,00E-03	1,50E-03	1,90E-03	3,07E-03	1,01E-03	2,87E-03	2,41E-03
	4,83E-03	5,50E-03	2,15E-03	2,04E-03	3,63E-03	1,73E-03	2,76E-03	2,66E-03
	6,17E-03	5,00E-03	1,84E-03	1,45E-03	3,09E-03	1,83E-03	2,26E-03	2,40E-03
B	4,38E-03	6,67E-03	1,91E-03	1,70E-03	3,17E-03	3,09E-03	1,62E-03	2,24E-03
	1,00E-02	4,33E-03	1,32E-03	1,33E-03	2,70E-03	3,73E-03	2,02E-03	2,02E-03
	8,17E-03	5,83E-03	1,01E-03	1,52E-03	2,96E-03	3,04E-03	1,46E-03	3,09E-03
C	4,33E-03	3,56E-03	2,10E-03	1,25E-03	2,39E-03	2,40E-03	1,57E-03	1,79E-03
	5,50E-03	2,67E-03	2,50E-03	1,50E-03	3,13E-03	1,68E-03	1,94E-03	1,62E-03
	6,83E-03	4,33E-03	2,42E-03	1,34E-03	2,46E-03	2,27E-03	1,87E-03	2,66E-03
Mean value	6,08E-03	4,88E-03	1,86E-03	1,56E-03	2,96E-03	2,31E-03	2,04E-03	2,32E-03
Median (Q2)	5,50E-03	5,00E-03	1,91E-03	1,50E-03	3,07E-03	2,27E-03	1,94E-03	2,40E-03
Q1	4,50E-03	4,33E-03	1,50E-03	1,34E-03	2,70E-03	1,73E-03	1,62E-03	2,02E-03
Q3	6,83E-03	5,83E-03	2,15E-03	1,70E-03	3,13E-03	3,04E-03	2,26E-03	2,66E-03
IQR	2,33E-03	1,50E-03	6,57E-04	3,62E-04	4,29E-04	1,30E-03	6,39E-04	6,33E-04
Upper	1,03E-02	8,08E-03	3,14E-03	2,25E-03	3,77E-03	4,99E-03	3,22E-03	3,61E-03
Lower	1,00E-03	2,08E-03	5,10E-04	8,00E-04	2,06E-03	-2,24E-04	6,62E-04	1,08E-03
% variation	-24,7%		-19,3%		-28,1%		12,0%	
DMSO								
	T = 28°C, IBU= 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l	
	Control	IBU	Control	IBU	Control	IBU	Control	IBU
A	2,19E-03	2,19E-03	2,00E-03	7,20E-03	2,19E-03	2,63E-03	2,00E-03	2,60E-03
	3,07E-03	1,53E-03	1,60E-03	6,60E-03	3,07E-03	1,31E-03	1,60E-03	2,20E-03
	2,63E-03	1,53E-03	1,40E-03	5,80E-03	2,63E-03	2,19E-03	1,40E-03	3,60E-03
B	1,53E-03	1,75E-03	6,00E-03	4,20E-03	1,53E-03	1,97E-03	6,00E-03	9,40E-03
	3,28E-03	1,75E-03	3,80E-03	7,00E-03	3,28E-03	1,09E-03	3,80E-03	3,60E-03
	1,53E-03	2,19E-03	7,00E-03	3,20E-03	1,53E-03	1,75E-03	7,00E-03	7,40E-03
C	2,41E-03	3,28E-03	3,60E-03	6,20E-03	2,41E-03	3,07E-03	3,60E-03	4,80E-03
	2,85E-03	2,63E-03	4,00E-03	5,40E-03	2,85E-03	2,63E-03	4,00E-03	4,60E-03
	2,63E-03	2,19E-03	3,20E-03	5,00E-03	2,63E-03	3,28E-03	3,20E-03	5,60E-03
Mean value	2,46E-03	2,12E-03	3,62E-03	5,62E-03	2,46E-03	2,21E-03	3,62E-03	4,87E-03
Median (Q2)	2,63E-03	2,19E-03	3,60E-03	5,80E-03	2,63E-03	2,19E-03	3,60E-03	4,60E-03
Q1	2,19E-03	1,75E-03	2,00E-03	5,00E-03	2,19E-03	1,75E-03	2,00E-03	3,60E-03
Q3	2,85E-03	2,19E-03	4,00E-03	6,60E-03	2,85E-03	2,63E-03	4,00E-03	5,60E-03
IQR	6,57E-04	4,38E-04	2,00E-03	1,60E-03	6,57E-04	8,76E-04	2,00E-03	2,00E-03
Upper	3,83E-03	2,85E-03	7,00E-03	9,00E-03	3,83E-03	3,94E-03	7,00E-03	8,60E-03
Lower	1,20E-03	1,09E-03	-1,00E-03	2,60E-03	1,20E-03	4,38E-04	-1,00E-03	6,00E-04
% variation	-16,1%		35,6%		-11,0%		25,6%	



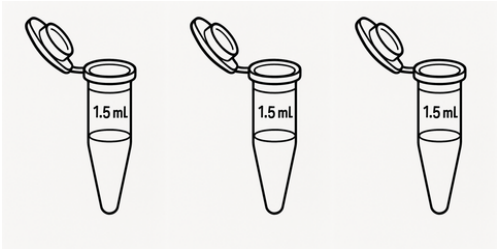
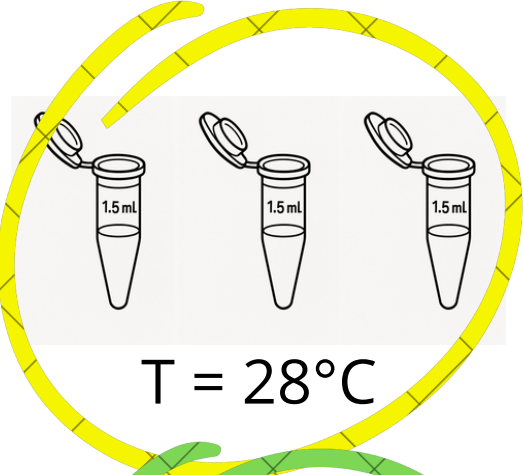
Confronto tra Controlli

Esperimento

Controllo

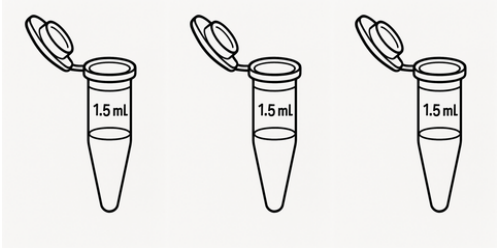
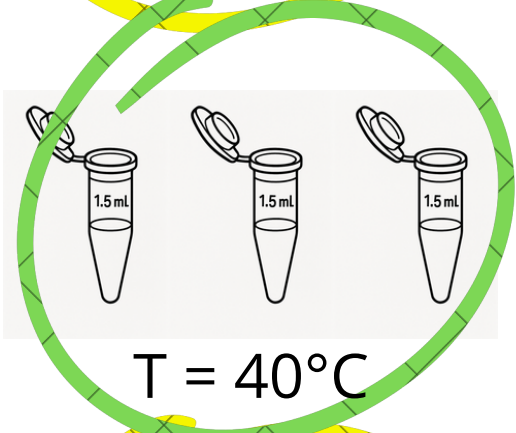
Condizione

1



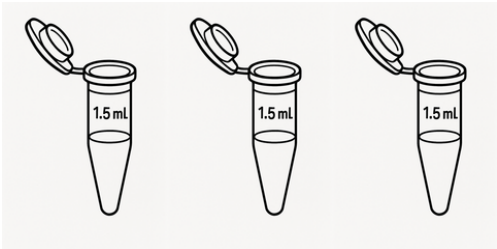
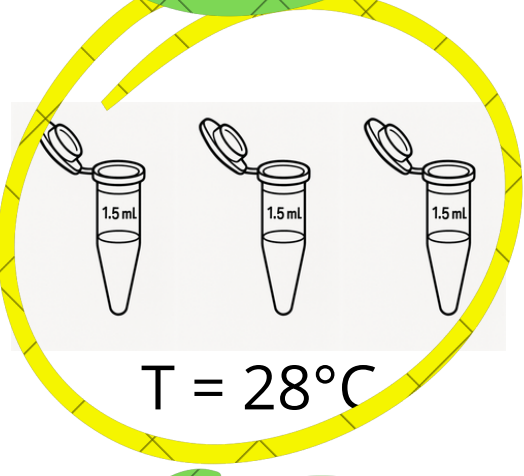
T = 28°C, IBU = 6.25 mg/l

2



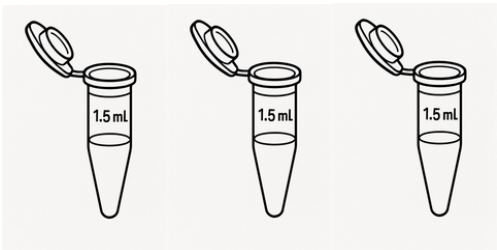
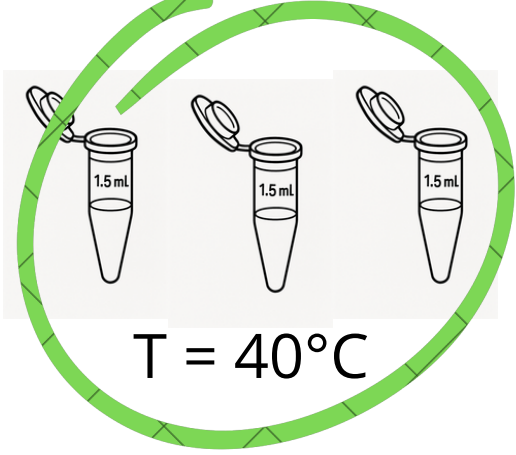
T = 40°C, IBU = 6.25 mg/l

3



T = 28°C, IBU = 50.0 mg/l

4



T = 40°C, IBU = 50.0 mg/l

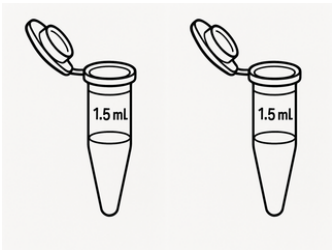
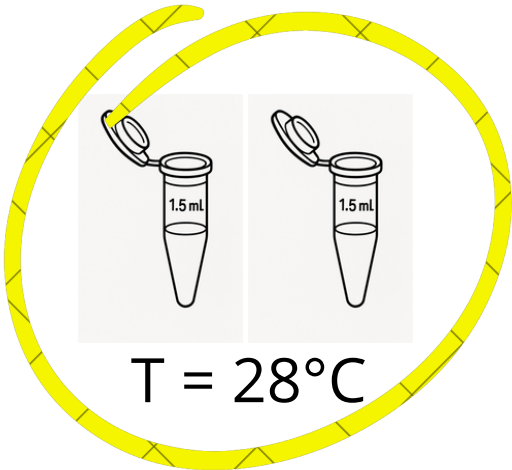
#	T [°C]	Ibuprofen	DMSO
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-

Esperimento

Controllo

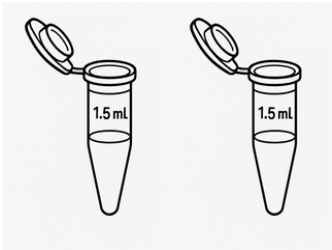
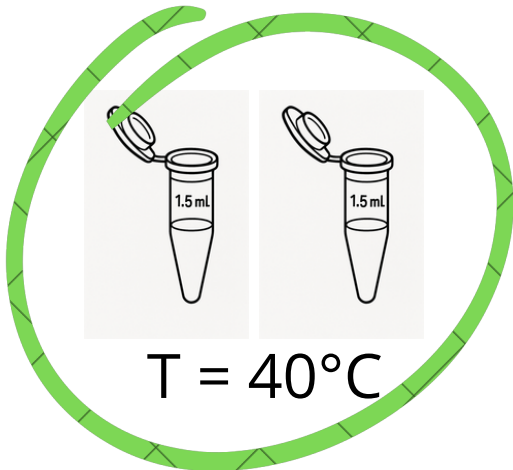
Condizione

9



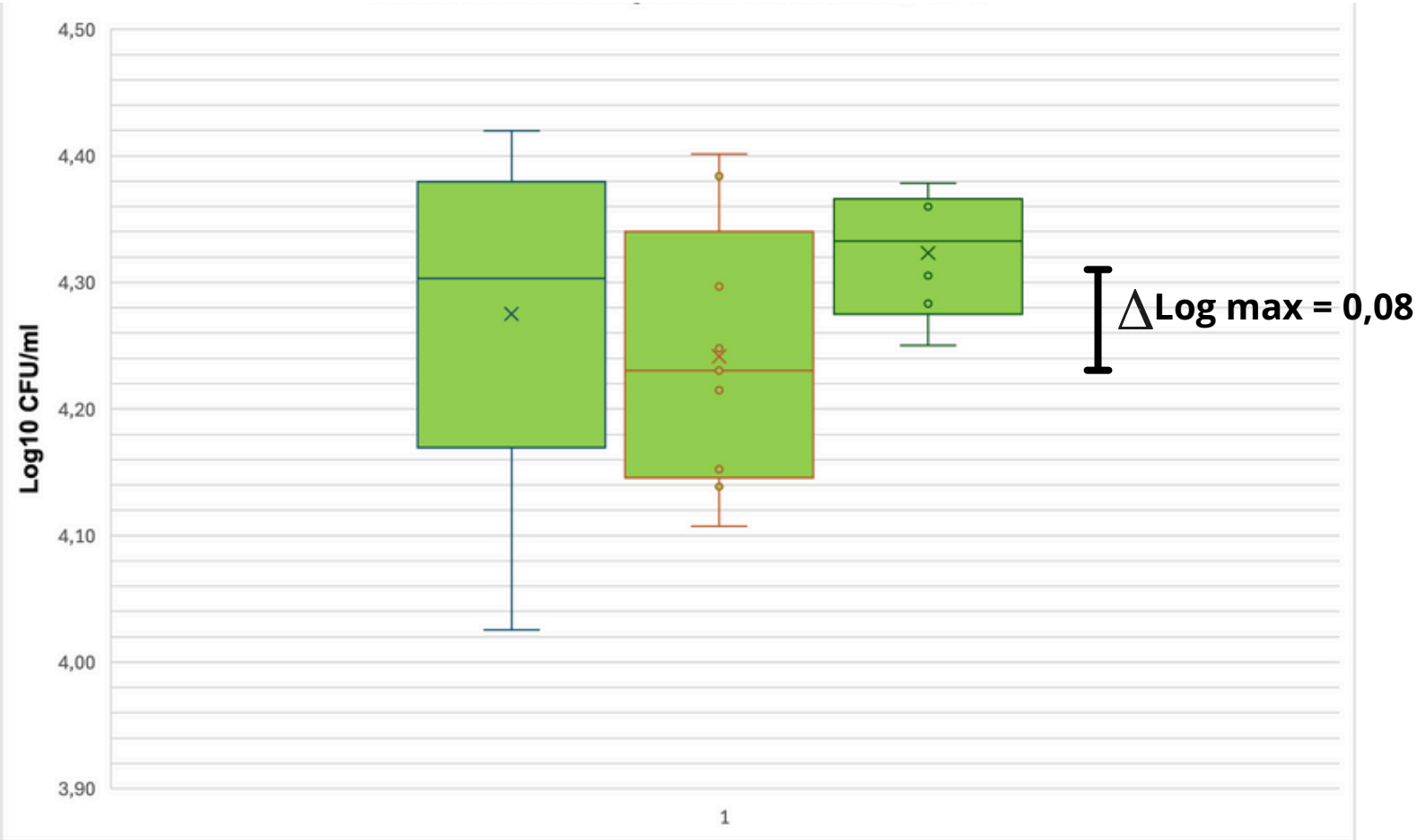
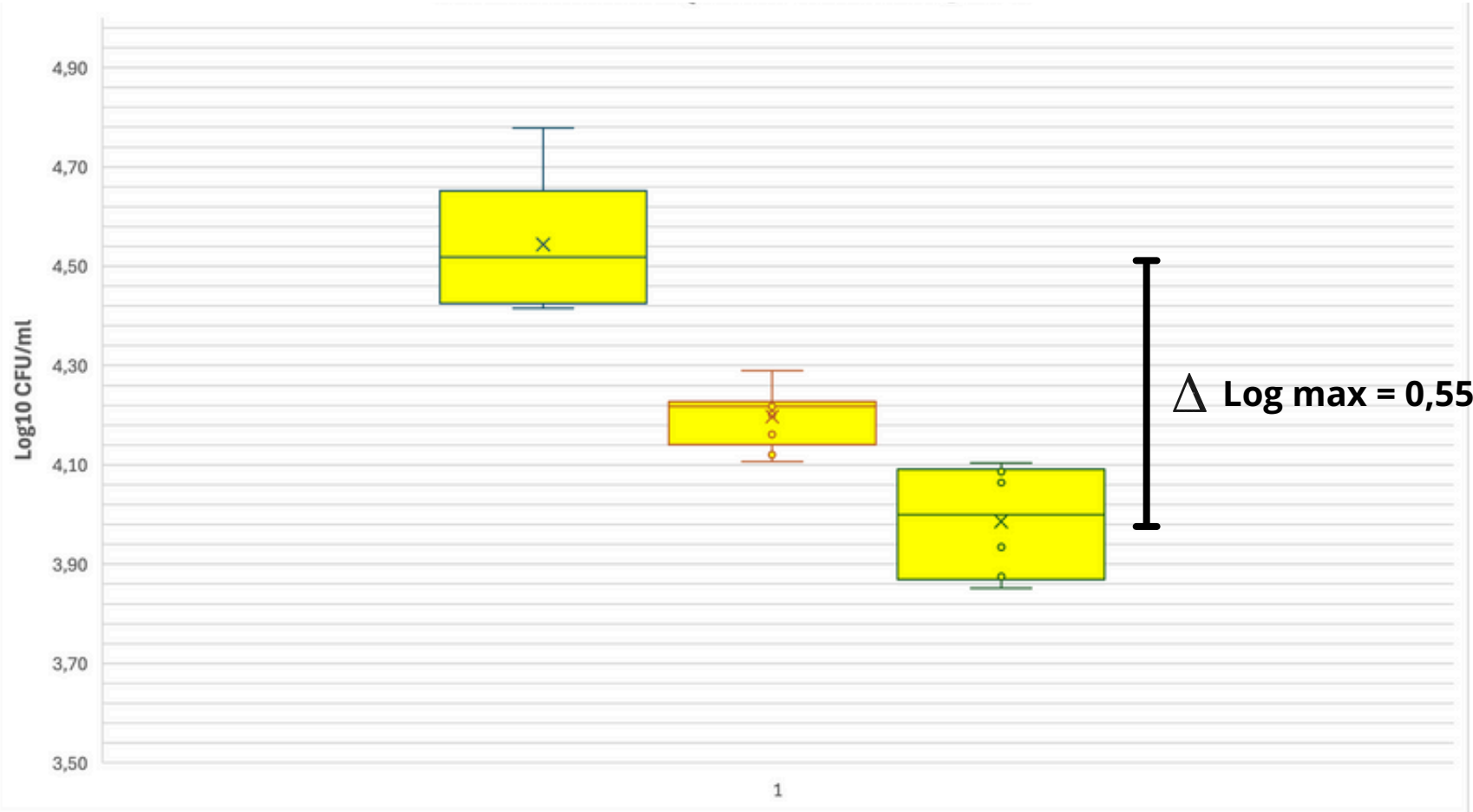
#	T [°C]	IBU [mg/l]	DMSO
9	-	0	+
10	+	0	+

10



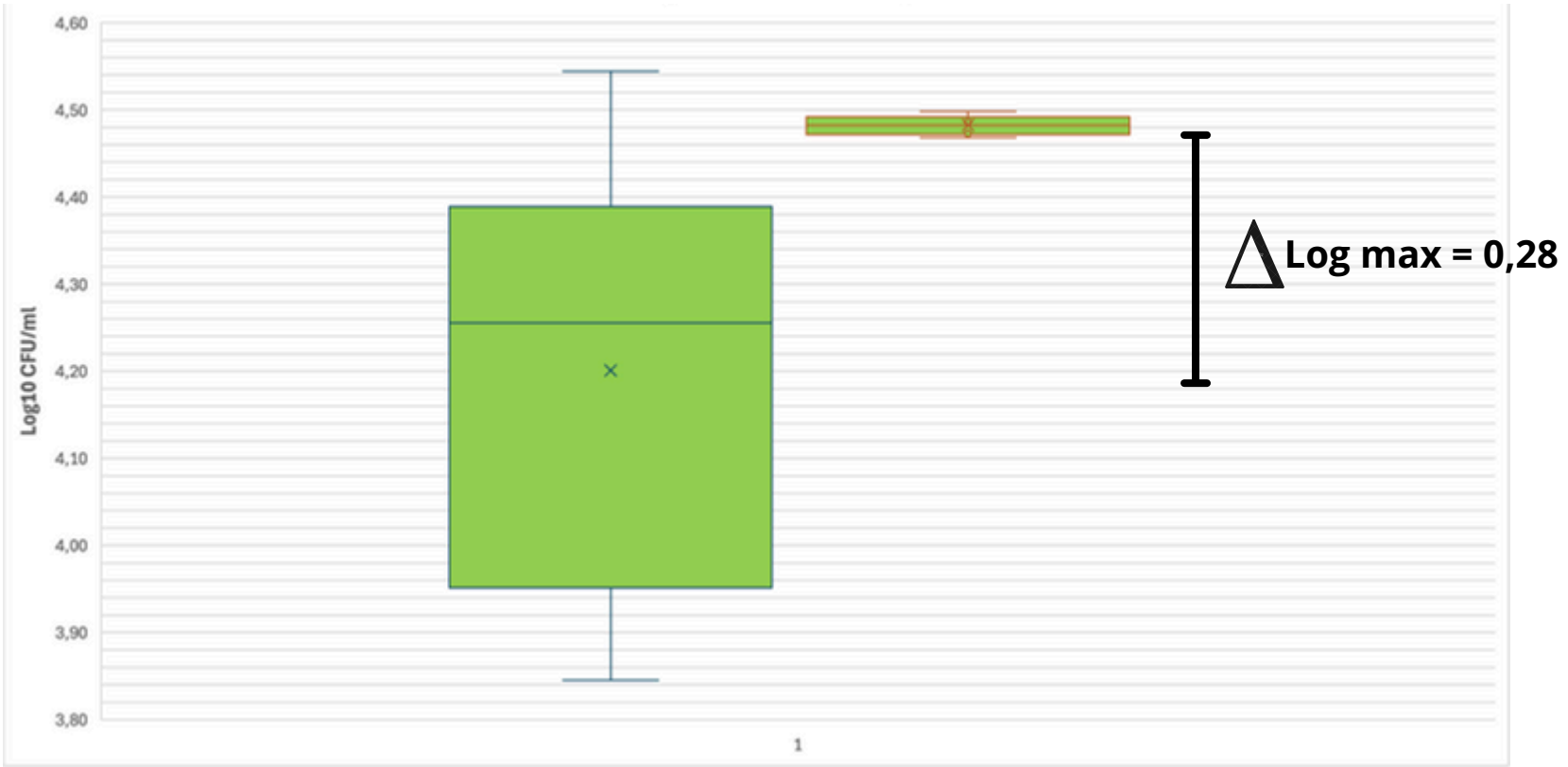
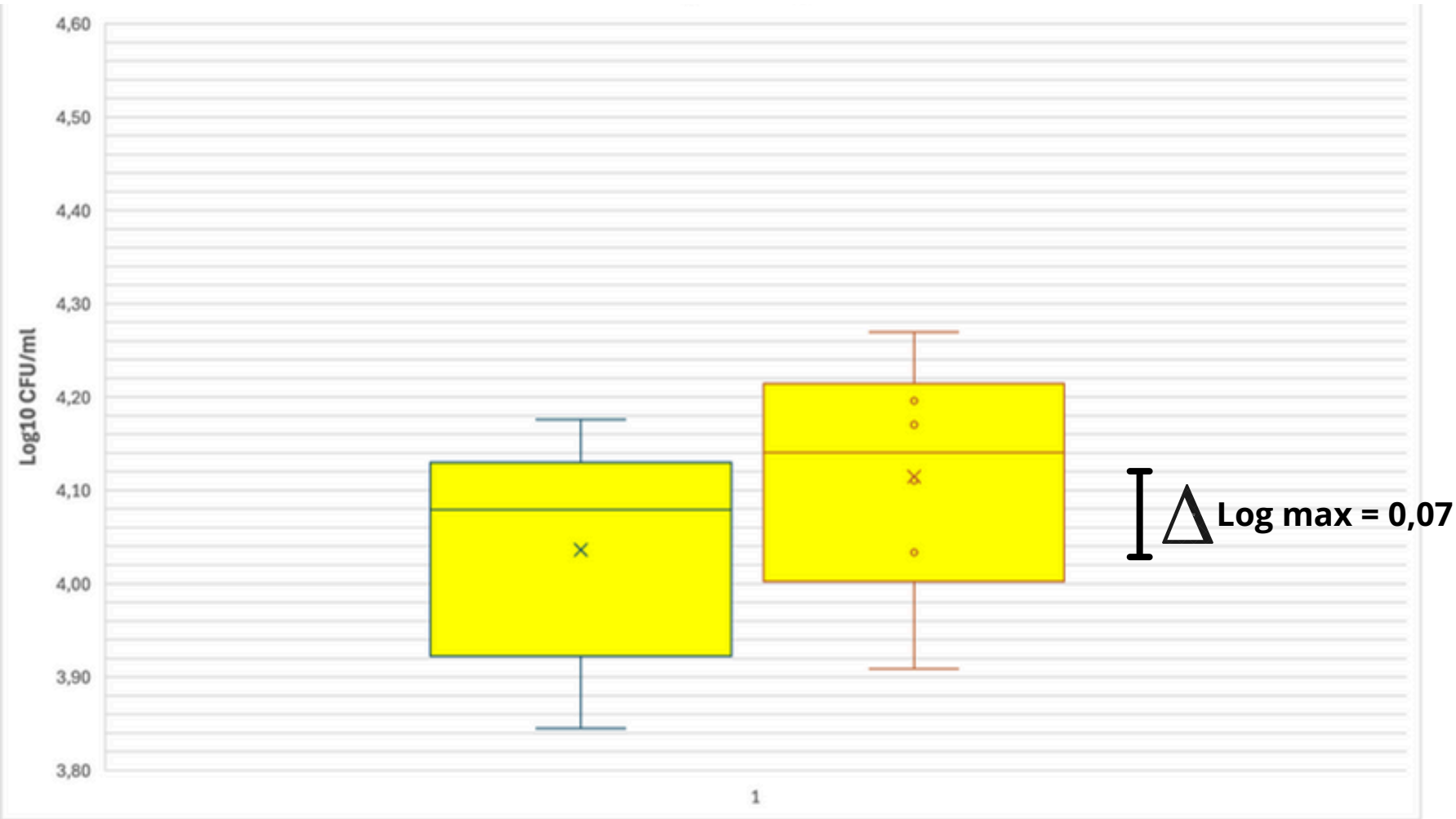
Confronto tra Controlli - NO DMSO

		NO DMSO								DMSO								DMSO vs NO DMSO			
		T=28°C, IBU= 6,25 mg/l		T=40°C, IBU= 6,25 mg/l		T=28°C, IBU= 50 mg/l		T=40°C, IBU= 50 mg/l		T=28°C, IBU= 6,25 mg/l		T=40°C, IBU= 6,25 mg/l		T=28°C, IBU= 50 mg/l		T=40°C, IBU= 50 mg/l		T=28°C		T=40°C	
		Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO
A		4,43	4,56	4,20	4,30	4,22	3,73	4,40	4,32	4,00	4,00	4,00	4,56	4,00	4,08	4,00	4,11	4,06	4,27	4,36	4,48
		4,46	4,52	4,35	4,33	4,29	3,97	4,38	4,37	4,15	3,85	3,90	4,52	4,15	3,78	3,90	4,04	4,09	4,20	4,28	4,49
		4,57	4,48	4,29	4,18	4,22	3,99	4,30	4,32	4,08	3,85	3,85	4,46	4,08	4,00	3,85	4,26	4,10	4,17	4,36	4,50
B		4,42	4,60	4,30	4,25	4,23	4,22	4,15	4,29	3,85	3,90	4,48	4,32	3,85	3,95	4,48	4,67	3,93	4,03	4,31	4,49
		4,78	4,41	4,14	4,15	4,16	4,30	4,25	4,25	4,18	3,90	4,28	4,54	4,18	3,70	4,28	4,26	3,88	3,91	4,38	4,47
		4,69	4,54	4,03	4,20	4,20	4,21	4,11	4,43	3,85	4,00	4,54	4,20	3,85	3,90	4,54	4,57	3,85	4,11	4,25	4,47
C		4,41	4,33	4,34	4,12	4,11	4,11	4,14	4,20	4,04	4,18	4,26	4,49	4,04	4,15	4,26	4,38				
		4,52	4,20	4,42	4,20	4,23	3,95	4,23	4,15	4,11	4,08	4,30	4,43	4,11	4,08	4,30	4,36				
		4,61	4,41	4,40	4,15	4,12	4,09	4,21	4,37	4,08	4,00	4,20	4,40	4,08	4,18	4,20	4,45				
Mean value		4,54	4,45	4,28	4,21	4,20	4,06	4,24	4,30	4,04	3,97	4,20	4,44	4,04	3,98	4,20	4,34	3,99	4,11	4,32	4,48
Standard deviation		0,13	0,13	0,13	0,07	0,06	0,17	0,10	0,09	0,12	0,11	0,24	0,11	0,12	0,16	0,24	0,20	0,11	0,13	0,05	0,01
Median		4,52	4,48	4,30	4,20	4,22	4,09	4,23	4,32	4,08	4,00	4,26	4,46	4,08	4,00	4,26	4,36	4,00	4,14	4,33	4,48
Δlog		-0,09		-0,07		-0,13		0,06		-0,06		0,24		-0,06		0,14		0,13		0,16	



Confronto tra Controlli - DMSO

		NO DMSO								DMSO								DMSO vs NO DMSO			
		T=28°C, IBU= 6,25 mg/l				T=40°C, IBU= 6,25 mg/l				T=28°C, IBU= 50 mg/l				T=40°C, IBU= 50 mg/l				T=28°C		T=40°C	
		Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO
A		4,43	4,56	4,20	4,30	4,22	3,73	4,40	4,32	4,00	4,00	4,00	4,56	4,00	4,08	4,00	4,11	4,06	4,27	4,36	4,48
		4,46	4,52	4,35	4,33	4,29	3,97	4,38	4,37	4,15	3,85	3,90	4,52	4,15	3,78	3,90	4,04	4,09	4,20	4,28	4,49
		4,57	4,48	4,29	4,18	4,22	3,99	4,30	4,32	4,08	3,85	3,85	4,46	4,08	4,00	3,85	4,26	4,10	4,17	4,36	4,50
B		4,42	4,60	4,30	4,25	4,23	4,22	4,15	4,29	3,85	3,90	4,48	4,32	3,85	3,95	4,48	4,67	3,93	4,03	4,31	4,49
		4,78	4,41	4,14	4,15	4,16	4,30	4,25	4,25	4,18	3,90	4,28	4,54	4,18	3,70	4,28	4,26	3,88	3,91	4,38	4,47
		4,69	4,54	4,03	4,20	4,20	4,21	4,11	4,43	3,85	4,00	4,54	4,20	3,85	3,90	4,54	4,57	3,85	4,11	4,25	4,47
C		4,41	4,33	4,34	4,12	4,11	4,11	4,14	4,20	4,04	4,18	4,26	4,49	4,04	4,15	4,26	4,38				
		4,52	4,20	4,42	4,20	4,23	3,95	4,23	4,15	4,11	4,08	4,30	4,43	4,11	4,08	4,30	4,36				
		4,61	4,41	4,40	4,15	4,12	4,09	4,21	4,37	4,08	4,00	4,20	4,40	4,08	4,18	4,20	4,45				
Mean value		4,54	4,45	4,28	4,21	4,20	4,06	4,24	4,30	4,04	3,97	4,20	4,44	4,04	3,96	4,20	4,34	3,99	4,11	4,32	4,48
Standard deviation		0,13	0,13	0,13	0,07	0,06	0,17	0,10	0,09	0,12	0,11	0,24	0,11	0,12	0,16	0,24	0,20	0,11	0,13	0,05	0,01
Median		4,52	4,48	4,30	4,20	4,22	4,09	4,23	4,32	4,08	4,00	4,26	4,46	4,08	4,00	4,26	4,36	4,00	4,14	4,33	4,48
Δlog		-0,09		-0,07		-0,13		0,06		-0,06		0,24		-0,06		0,14		0,13		0,16	



Risultati - riassunto

- Efficienza di coniugazione alta (90-100%) sui campioni selezionati.
- IncN passa sempre mentre incF praticamente mai, solo qualche caso isolato a 40°C e in assenza di ibuprofene e DMSO.
- Ibuprofene (in assenza di DMSO) diminuisce sempre efficienza di coniugazione, ad eccezione di una condizione in cui si osserva un leggero aumento (40°C, IBU 50.0mg/l).
- DMSO influisce positivamente sulla crescita dei transconiuganti, sia in assenza che in presenza di ibuprofene, in particolar modo a 40°C. Effetto più limitato a 28°C.
- Interazione tra ibuprofene e DMSO positiva per la crescita a 40°C.

Gestione dati di conta batterica?

True CFU_100ul

			NO DMSO								DMSO								DMSO vs NO DMSO					
			T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C		T = 40°C			
			Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO		
Condition Replicate	A	-1	DMTC	DMTC	157	200	165	54	252	211	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	116	186	229	301
			DMTC	DMTC	226	214	195	93	242	233	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	122	157	192	309
			DMTC	DMTC	192,9	152,6	166	96	196	210	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	127	148	230	315
		-2	27	36	4	5	16	6	15	12	10	10	10	26	10	12	10	13	9	1	3	1	1	
			29	33	7	2	13	10	11	5	14	7	8	33	14	6	8	11	13	4	3	3	3	
			37	30	11	3	14	7	13	3	12	7	7	29	12	10	7	18	8	5	5	3	3	
	-3	3	4	0	1	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
		7	6	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
		3	19	0	0	1	0	2	0	1	1	1	0	0	0	1	5	0	0	0	0	0		
	B	-1	DMTC	DMTC	201	179	170	165	142	196	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	66	108	202	306
			DMTC	DMTC	139	140	145	200	177	177,5	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	75	81	239	297
			DMTC	DMTC	106	160	159	163	138	271	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	71	129	178	294
		-2	26,25	40	10	7	13	23	9	5	7	8	30	21	7	9	30	47	1	3	3	3		
			60	26	7	9	11	18	10	11	15	8	19	35	15	5	19	18	5	5	3	7		
			49	35	6	8	10	17	14	16	7	10	35	16	7	8	35	37	4	2	4	8		
	-3	4	10	0	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	1			
		3	5	3	3	1	2	0	1	1	1	0	0	0	0	0	3	0	0	0	0			
		2	6	0	2	2	2	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0		
C	-1	DMTC	DMTC	221	131	128	129	137,6	157	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC							
		DMTC	DMTC	263	157	168	90	170	142	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC							
		DMTC	DMTC	254	141	132	122	164	233	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC	DMTC							
	-2	26	21,3	15	18	3	16	4	3	11	15	18	31	11	14	18	24							
		33	16	9	15	4	8	3	5	13	12	20	27	13	12	20	23							
		41	26	11	11	5	8	5	6	12	10	16	25	12	15	16	28							
-3	8	3	2	1	0	1	0	0	0	1	0	0	0	0	0	2								
	5	3	0	0	0	1	0	0	0	1	0	1	0	0	1	1								
	25	6	1	1	1	0	0	0	1	0	0	0	0	0	0	0								

True CFU/ml

	NO DMSO								DMSO								DMSO vs NO DMSO			
	T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C, IBU = 6,25 mg/l		T = 40°C, IBU = 6,25 mg/l		T = 28°C, IBU = 50 mg/l		T = 40°C, IBU = 50 mg/l		T = 28°C		T = 40°C	
	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	IBU	Control	DMSO	Control	DMSO
A	27000	36000	11700	20000	16500	5400	25200	21100	10000	10000	10000	36000	10000	12000	10000	13000	11600	18600	22900	30100
	29000	33000	22600	21400	19500	9300	24200	23300	14000	7000	8000	33000	14000	6000	8000	11000	12200	15700	19200	30900
	37000	30000	19290	15260	16600	9600	19600	21000	12000	7000	7000	29000	12000	10000	7000	18000	12700	14800	23000	31500
B	26250	40000	20100	17900	17000	16500	14200	19600	7000	8000	30000	21000	7000	9000	30000	47000	6600	10800	20200	30600
	60000	26000	13900	14000	14500	20000	17700	17750	15000	8000	19000	35000	15000	5000	19000	18000	7500	8100	23900	29700
	49000	35000	10600	16000	15900	16300	12800	27100	7000	10000	30000	16000	7000	8000	35000	37000	7100	12900	17800	29400
C	26000	21300	22100	13100	12800	12900	13760	15700	11000	15000	18000	31000	11000	14000	18000	24000				
	33000	16000	26300	15700	16800	9000	17000	14200	13000	12000	20000	27000	13000	12000	20000	23000				
	41000	26000	25400	14100	13200	12200	16400	23300	12000	10000	16000	25000	12000	15000	16000	28000				
Mean value	36472	29256	19554	16364	15867	12378	17896	20339	11222	9667	18111	28111	11222	10111	18111	24333	9950	13483	21167	30367
Standard deviation	11742	7654	5277	2831	2082	4569	4423	4042	2819	2508	9532	6623	2819	3444	9532	11597	2502	3724	2448	764

